DIR-HSI: Degradation-Invariant Representation Learning for Hyperspectral Image Classification

Boxiang Yang, Ning Chen, Yingbo Fan, Zexi Shi, Haoyuan Zhang, Hui Liu, Run Niu, and Shanjun Mao

***Abstract*—In recent years, hyperspectral image (HSI) classification has achieved remarkable progress in fine-grained surface recognition and intelligent remote sensing analysis. However, real-world degradations—such as sensor noise, compression artifacts, and atmospheric interference—often introduce severe domain shifts, leading to degraded robustness and limited generalization of existing models. To address this challenge, we propose DIR-HSI, a degradation-invariant hyperspectral classification framework designed to enhance feature stability and discriminability under multidegradation conditions. The framework comprises two key modules: the degradation-invariant representation by Fourier-domain feature extraction (DIR-FFE) module, which leverages Fourier-domain feature decomposition to extract discriminative and degradation-insensitive frequency features, and the degradation-invariant representation contrastive enhancement (DIR-CE) module, which employs cross-degradation supervised contrastive (SupCon) learning to enforce consistency among same-class representations across different degradation conditions. Together, these components enable effective disentanglement of degradation-sensitive and stable factors, maintaining feature consistency and reinforcing class separability. The key advantage of DIR-HSI lies in its ability to jointly model degradation-invariant and discriminative representations, achieving robust and generalizable performance across complex degradation scenarios. Extensive experiments on multiple benchmark datasets demonstrate that DIR-HSI consistently outperforms state-of-the-art methods in terms of accuracy, robustness, and generalization. The code will be available at https://github.com/yangboxiang1207/DIR-HSI**

***Index Terms*—Contrastive learning, degradation invariance, Fourier-domain feature, hyperspectral image (HSI) classification.**

Received 29 December 2025; revised 6 March 2026 and 21 April 2026; accepted 23 April 2026. Date of publication 27 April 2026; date of current version 4 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62505008, in part by the Postdoctoral Research Foundation of China under Grant 2024M751474, and in part by the Project of Sihong County Active Fault Detection and Earthquake Hazard Assessment under Grant 2023h659. *(Corresponding author: Ning Chen.)*

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Digital Object Identifier 10.1109/TGRS.2026.3687650

I. Introduction

HYPERSPECTRAL imaging, as an important branch of remote sensing technology, enables the acquisition of fine-grained surface reflectance characteristics across continuous and narrow spectral bands, thereby providing significantly higher spectral resolution and richer information than traditional multispectral imaging [1]. Owing to its exceptional capability in fine-grained material classification and object recognition, hyperspectral image (HSI) has been widely applied in various fields such as precision agriculture [1], [2], mineral exploration [3], urban planning [4], ecological and environmental monitoring [5], and disaster assessment [6], [7], becoming a key technology for advancing the intelligent development of Earth observation. In recent years, with the continuous improvement of sensor resolution and imaging efficiency, HSI-based object classification and semantic understanding have played an increasingly important role in intelligent remote sensing systems [8].

However, hyperspectral data are highly susceptible to multisource degradations during acquisition, transmission, and preprocessing [9], including sensor noise, illumination variation, compression artifacts, atmospheric scattering, blurring, and striping distortions [10]. These degradations not only disrupt the spectral continuity and spatial structural integrity of the data but also cause significant domain shifts [11]. For instance, thermal noise and electronic interference often introduce zero-mean Gaussian noise [12], lens defocus, and transmission loss lead to spatial blurring, while atmospheric scattering and water vapor absorption alter the reflectance distribution across wavelengths [13]. Such complex and interdependent degradations frequently coexist in real-world conditions, making it challenging for HSI classification models to maintain both feature consistency and discriminative capability under degraded scenarios. Consequently, enhancing the robustness and cross-domain generalization of HSI classification models under multiple degradation conditions has emerged as a critical challenge in the field of intelligent hyperspectral analysis.

With the rapid advancement of deep learning in intelligent remote sensing, HSI classification has evolved from handcrafted feature engineering to deep representation learning [14]. Early approaches primarily relied on traditional machine learning models [15], such as support vector machines (SVMs) [16], [17], random forests (RFs) [18], k-nearest neighbors (K-NNs) [19], and principal component analysis (PCA) [20]. The introduction of convolutional neural networks (CNNs)

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marked a major shift toward joint spectral–spatial modeling [21], [22]. Representative architectures, including 3D-CNNs [23], [24], [25], ResNet [26], [27], and spectral–spatial residual networks (SSRNs) [28], employ local convolutional kernels to capture neighborhood context across spectral and spatial dimensions. Despite their effectiveness, CNN-based methods are constrained by limited receptive fields and often exhibit degraded robustness under severe noise or distortions [29]. To model sequential spectral dependencies, recurrent neural networks (RNNs) [30] and long short-term memory (LSTM) [31] networks were further explored; however, their training inefficiency on high-dimensional spectra and limited spatial modeling capability restrict their practical applicability. More recently, Transformer-based architectures have emerged as a powerful alternative for HSI classification, leveraging self-attention mechanisms to capture long-range spectral–spatial dependencies [32], [33], [34], [35]. Recent studies have also explored more advanced attention and meta-learning strategies to enhance representation capability and adaptability in HSI classification, such as the hybrid attention network with deterministic–probabilistic mechanisms [36] and cross-domain meta-learning frameworks designed for few-shot scenarios [37]. Nevertheless, most existing Transformer-based HSI models directly adopt generic architectures originally designed for natural images, lacking task-specific inductive biases, which leads to strong data dependence and suboptimal training stability under limited samples or severe degradations [38].

In parallel, recent years have witnessed rapid progress in hyperspectral large models and related representation learning paradigms, driven by large-scale pretraining and self-supervised learning across diverse scenes, sensors, and tasks [39]. By leveraging massive unlabeled data, these approaches aim to learn generalizable spectral–spatial representations that can be transferred to downstream hyperspectral analysis tasks, thereby reducing annotation dependence [40], [41]. However, their primary emphasis lies in cross-task generalization rather than robustness under complex degradation conditions. In practice, most existing hyperspectral large models do not explicitly model degradation mechanisms or enforce representation stability across diverse corruption types [42], making them vulnerable to severe noise, spatial blurring, and atmospheric interference. Meanwhile, lightweight and state-space modeling approaches, such as Mamba [14], [43] and RetNet [44], [45], have been introduced to improve modeling efficiency while capturing long-range dependencies. Although computationally efficient, these methods still rely heavily on high-quality inputs and exhibit limited robustness under multisource degradations or cross-domain shifts, where learned representations tend to become unstable and inconsistent.

To mitigate degradation-induced performance drops, recent studies have explored various strategies, including noise suppression [46], feature reconstruction [47], data augmentation [48], and contrastive learning [49]. In particular, several efforts have begun to address hyperspectral classification under noisy or degraded conditions more explicitly. For example, seeing through the noise [50] introduces a cross-modal guided classification framework that leverages complementary active remote sensing data such as LiDAR to compensate for degraded hyperspectral observations, and further enhances robustness through hierarchical cross-modal guidance and degradation-aware feature mixing. S3CFSL [51] improves cross-domain few-shot hyperspectral classification by incorporating feature denoising operations into the feature extraction process and coupling them with a semantic-aware domain alignment strategy, thereby reducing the influence of residual noise and nonessential information on transferable learning. In addition, HyperTTA [52] formulates degraded hyperspectral classification as a distribution-shift problem, constructs a multidegradation benchmark with nine representative corruption types, and improves robustness through source-free test-time adaptation on unlabeled degraded target samples. These studies demonstrate the growing importance of noisy-sample-oriented hyperspectral analysis. Nevertheless, existing methods are still largely tailored to specific robustness settings, such as cross-modal compensation, denoising-assisted few-shot transfer, or test-time adaptation, and generally lack a unified and systematic mechanism for explicitly modeling degradation-sensitive versus degradation-invariant features. As a result, performance may remain inconsistent under complex and compound degradation scenarios commonly encountered in real-world hyperspectral imagery.

Against this backdrop, frequency-domain modeling has emerged as a promising direction for enhancing model robustness under degraded conditions. Prior studies have shown that many degradation processes—such as random noise, blurring, and compression artifacts—exhibit more explicit and structured manifestations in the frequency domain, often perturbing specific frequency components and altering energy distributions [53]. As a result, frequency-domain representation learning has been widely recognized as an effective strategy for improving robustness to degradation. However, it is important to note that frequency components cannot be simplistically interpreted from a binary perspective. In hyperspectral imagery, fine-grained spatial textures, structural boundaries, and subtle material patterns—especially those critical for discriminating spectrally similar crops or surface materials—also reside in middle- and high-frequency regions. Treating high-frequency components as pure degradation noise and suppressing them indiscriminately may therefore lead to the loss of essential discriminative information. From this perspective, the core challenge of frequency-domain modeling lies not in statically separating low- and high-frequency bands, but in identifying and modeling frequency components whose responses are unstable under different degradation conditions, degradation-sensitive components, while preserving those that remain stable and discriminative. Although several studies [54], [55], [56] have introduced frequency-domain modeling into HSI classification tasks, most of these methods were not systematically designed for degradation scenarios. They often rely on fixed filters or predefined frequency selections, lacking a learnable spectral decomposition mechanism. Such limitations hinder the model's ability to adaptively distinguish between degradation-sensitive and degradation-invariant features across varying degradation conditions. Consequently, while these methods can suppress certain noise patterns, they fail to systematically capture the intrinsic relationship between

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degradation perturbations and representation stability at the feature level.

Based on the above analysis, the key challenge in hyperspectral remote sensing under multidegradation scenarios lies in maintaining the stability of feature representations amid complex and variable degradation conditions, thereby achieving steady characterization across diverse degradation types. Most existing approaches primarily focus on spatial or spectral noise suppression while overlooking the statistical and structural properties of degradation in the frequency domain. As a result, they fail to extract stable components that are insensitive to degradation perturbations, exacerbating domain shifts and weakening model generalization. Therefore, it is essential to construct a degradation-invariant representation in the frequency domain and to explicitly disentangle degradation-sensitive and degradation-stable features through an end-to-end learnable mechanism, in order to enhance model robustness and cross-domain adaptability under complex degradation conditions.

To this end, this article proposes a DIR-HSI classification framework, which systematically addresses the insufficient modeling of degradation features from the joint perspective of frequency-domain representation learning and discriminative feature optimization. Specifically, the degradation-invariant representation by Fourier-domain feature extraction (DIR-FFE) module performs Fourier-domain decomposition and adaptive spectral suppression to effectively separate degradation-sensitive components from stable ones. Meanwhile, the degradation-invariant representation contrastive enhancement (DIR-CE) module introduces a cross-degradation contrastive learning constraint that enforces feature consistency among samples of the same class under various degradation conditions, while simultaneously enlarging interclass margins. The two modules work in synergy—ensuring stability along the degradation dimension and strong discriminability along the class dimension—thus systematically mitigating domain shift across complex degradation scenarios and substantially enhancing the robustness and generalization capability of HSI classification. The primary contributions of our approach can be summarized in three aspects.

1) This article proposes an HSI classification framework, DIR-HSI, designed for multidegradation scenarios. The framework constructs a degradation-invariant representation and leverages contrastive learning to further enhance the consistency and discriminability of features under varying degradation conditions, thereby enabling robust classification of degraded hyperspectral data. Experimental results demonstrate that DIR-HSI consistently outperforms existing methods across multiple public datasets, exhibiting superior generalization capability.
2) To address the domain shift issue commonly observed in HSI under multidegradation conditions, this article proposes the DIR-FFE module. By leveraging Fourier-domain feature decomposition, the proposed module extracts discriminative and degradation-insensitive stable features in the frequency space, thereby enhancing the model's classification robustness and cross-domain generalization capability in degraded scenarios.
3) To further enhance the discriminative capability of the degradation-invariant representation, a contrastive learning mechanism is incorporated (DIR-CE). During training, multiple degraded variants are generated for each sample to construct cross-degradation contrastive constraints, encouraging the model to draw the representations of the same sample under different degradation conditions closer while enlarging the interclass feature distinctions. This process effectively separates degradation-sensitive and degradation-stable components, thereby reinforcing the robustness and discriminability of the degradation-invariant features and ultimately improving the model's generalization performance.

II. Methodology

In this section, we introduce the proposed DIR-HSI classification framework, which is designed to address the challenges of domain shift and robustness degradation commonly encountered in HSI classification under multidegradation conditions.

A. Overview

As illustrated in Fig. 1, the model mainly consists of two core components: the DIR-FFE module and the DIR-CE module, which collaboratively achieve degradation-invariant feature extraction and cross-degradation representation alignment.

The DIR-FFE module divides the hyperspectral data into spatial patches and applies a channel-wise 2-D Fourier transform (2D-FFT) to map them into the frequency domain, separating high-frequency noise from low-frequency structural information. An adaptive spectral suppression module (ASSM) then performs learnable low-pass filtering to suppress degradation noise while preserving key spectral structures. The filtered frequency features are fused with spatial features and fed into the TransformerNet backbone for joint spectral–spatial representation learning.

The DIR-CE module adopts a cross-degradation supervised contrastive (SupCon) strategy to enhance feature consistency and discriminability. Clean and degraded embeddings form positive and negative pairs based on class labels, optimized by the SupCon Loss, which clusters same-class features (clean and degraded) while separating different classes. This process disentangles degradation-sensitive and invariant components, yielding compact intraclass features and clearer interclass boundaries.

In general, DIR-HSI integrates robust Fourier-domain feature modeling (DIR-FFE) with cross-degradation contrastive optimization (DIR-CE) to jointly achieve degradation invariance and discriminative enhancement. This collaborative design enables the framework to suppress degradation-induced distortions, maintain representation consistency across diverse degradation conditions, and significantly improve classification robustness and generalization performance under complex real-world scenarios.

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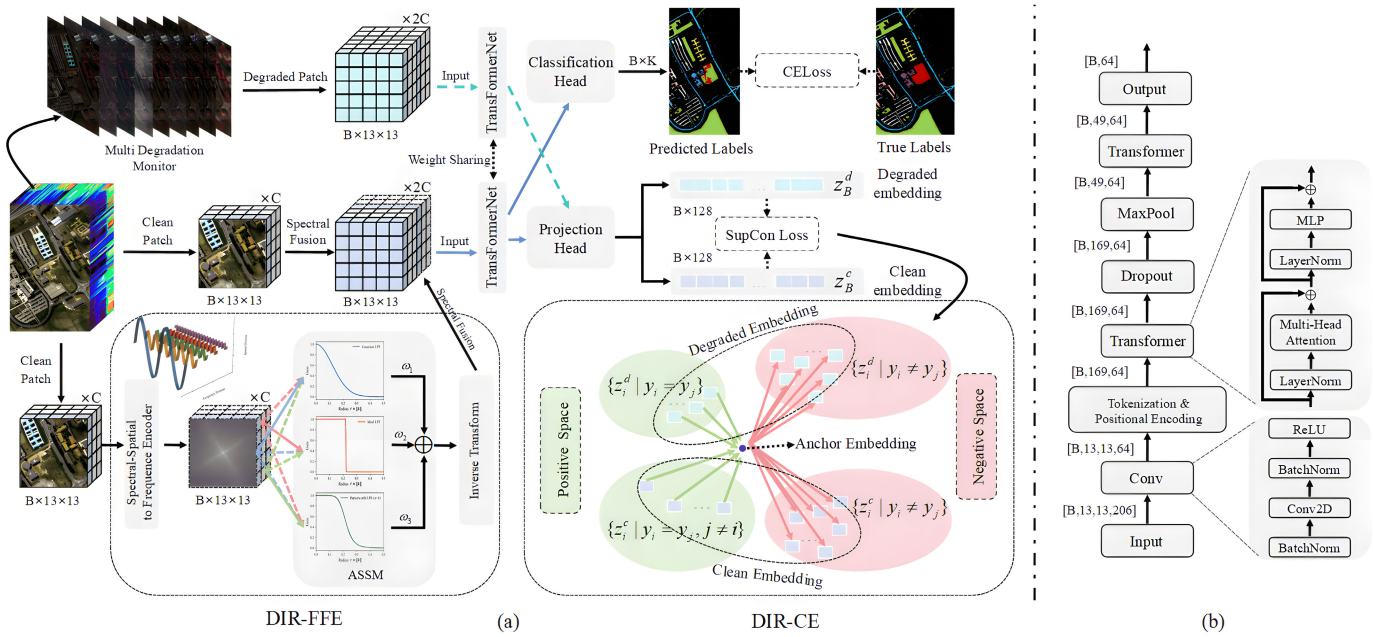


Fig. 1. Overview of the proposed DIR-HSI. (a) Architecture of the DIR-HSI classification framework consisting of the DIR-FFE and DIR-CE modules. B denotes the batch size used during training and K denotes the number of classes in the classification task. (b) Architecture of the TransformerNet and its input–output data flow. For clarity of illustration, the spectral dimension C is instantiated using the PU dataset as an example, where $C = 103$.

B. DIR by Fourier-Domain Feature Extraction

The DIR-FFE is designed to extract degradation-invariant and spectrally stable representations by leveraging frequency-domain feature decomposition. Unlike conventional spatial-domain convolutional or transformer-based models that are highly sensitive to degradation-induced distortions (e.g., noise, blur, and compression artifacts), DIR-FFE performs spectral–frequency encoding and adaptive spectral filtering to separate stable low-frequency structures from degradation-sensitive high-frequency components. This process enables the model to maintain discriminative and consistent representations under diverse degradation conditions.

1) Input Representation and Patch Construction: To enable frequency-domain modeling while preserving local spatial context, the DIR-FFE processes HSI in a patch-wise manner. Given an input HSI $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, it is partitioned into a set of spatial patches, as shown in the following equation:

$$\mathbf{P}_i \in \mathbb{R}^{P \times P \times C}, \quad i = 1, 2, \ldots, N \tag{1}$$

where P denotes the patch size, C is the number of spectral bands, and N is the total number of extracted patches.

Within each patch, spectral bands are treated independently, and the spatial content of each band is processed in a channel-wise manner. This formulation provides a unified spectral–spatial input representation for subsequent frequency-domain transformation, without introducing additional spatial structure modeling at this stage. Such a design keeps the input representation compact and avoids ambiguity between frequency-domain feature extraction and spatial relationship learning, which is handled later by the backbone network.

2) Spectral–Spatial to Frequency Encoder: Given a spatial patch $\mathbf{P}_i \in \mathbb{R}^{P \times P \times C}$, the DIR-FFE maps spectral–spatial information into the frequency domain via a channel-wise spatial Fourier transform. For each spectral band $c \in \{1, \ldots, C\}$, the corresponding spatial slice $\mathbf{P}_i(:, :, c)$ is independently transformed using a 2-D discrete Fourier transform (2D-DFT), defined as shown in the following equation:

$$\mathbf{F}_i(u, v, c) = \sum_{x=0}^{P-1} \sum_{y=0}^{P-1} \mathbf{P}_i(x, y, c) \exp\left(-j2\pi \frac{ux + vy}{P}\right) \tag{2}$$

where (u, v) denote the frequency coordinates.

The frequency-domain representation $\mathbf{F}_i \in \mathbb{C}^{P \times P \times C}$ has the same spatial resolution as the input patch along the frequency axes. This dimensional correspondence follows directly from the definition of the 2D-DFT on finite spatial grids and does not introduce any additional modeling assumptions. Such a transformation functions as a spectral–spatial frequency encoder, exposing degradation-sensitive and degradation-stable components in the frequency domain, while explicit spatial structure modeling is deferred to the backbone network.

3) Adaptive Spectral Suppression Module: The ASSM constitutes the core component of DIR-FFE, aiming to dynamically suppress degradation-sensitive spectral components while preserving discriminative information in the frequency domain. Unlike fixed low-pass filtering strategies, ASSM adopts parameterized attenuation kernels with learnable fusion, enabling data-driven adaptation to diverse degradation patterns.

Given the frequency-domain representation $\mathbf{F}_i$ obtained in SS2F, ASSM first constructs three candidate attenuation responses using parameterized low-pass kernels. Under a centered frequency convention, the radial frequency distance is defined as shown in the following equation:

$$D(u, v) = \sqrt{(u - u_0)^2 + (v - v_0)^2}, \quad u_0 = v_0 = \left\lfloor \frac{P}{2} \right\rfloor \tag{3}$$

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where (u_0, v_0) denotes the zero-frequency (direct current) component at the center of the frequency plane.

Based on this definition, three complementary attenuation kernels are considered.

1) Gaussian kernel, as shown in the following equation:

$$H_{\text{gauss}}(u, v; \sigma) = \exp\left(-\frac{D(u, v)^2}{2\sigma^2}\right) \tag{4}$$

where σ controls the spectral attenuation scale.

2) Ideal low-pass kernel, as shown in the following equation:

$$H_{\text{ideal}}(u, v; \rho) = \frac{1}{1 + \exp(\beta(D(u, v) - \rho))} \tag{5}$$

where ρ denotes the cutoff radius and β controls the smoothness of the transition.

3) Butterworth low-pass kernel, as shown in the following equation:

$$H_{\text{butter}}(u, v; \rho, n) = \frac{1}{1 + \left(\frac{D(u,v)}{\rho}\right)^{2n}} \tag{6}$$

where ρ is the cutoff radius and n is the filter order.

Here, ρ and σ play distinct roles. ρ determines the cutoff boundary in the frequency plane, while σ controls the decay rate of Gaussian attenuation. This formulation resolves the ambiguity between variance-related parameters and cutoff-related parameters.

Each kernel is applied in parallel to the frequency-domain patch, as shown in the following equation:

$$\mathbf{G}_i^{(k)}(u, v, c) = H_k(u, v) \odot \mathbf{F}_i(u, v, c), \quad k \in \{1, 2, 3\} \tag{7}$$

where $\odot$ denotes the element-wise multiplication.

Rather than selecting a single kernel, ASSM employs a learnable gated fusion mechanism to interpolate among the three filtered responses, as shown in the following equation:

$$\tilde{\mathbf{F}}_i = \sum_{k=1}^{3} \omega_k \mathbf{G}_i^{(k)}, \quad \boldsymbol{\omega} = \text{Softmax}(\mathbf{w}) \tag{8}$$

where $\omega = (\omega_1, \omega_2, \omega_3)$ are learnable fusion weights satisfying $\sum_k \omega_k = 1$. This soft selection strategy ensures differentiability and allows ASSM to adaptively emphasize different attenuation behaviors depending on the degradation characteristics and scene content. The kernel parameters are jointly denoted as shown in the following equation:

$$\alpha = (\rho, \sigma, n) \tag{9}$$

which are learned end-to-end during training. For efficiency and stability, the kernel parameters are shared across spectral channels within each layer. To ensure numerical stability and meaningful spectral responses, these parameters are constrained within predefined ranges via bounded reparameterization.

The fused frequency-domain representation $\widehat{\mathbf{F}}_i$ is transformed back to the spatial domain using an inverse Fourier transform, yielding a spectrally suppressed spatial patch $\hat{\mathbf{P}}_i \in \mathbb{R}^{P \times P \times C}$. Instead of replacing the original patch, ASSM adopts a residual fusion strategy by concatenating the original spatial patch $\mathbf{P}_i$ and the refined patch $\hat{\mathbf{P}}_i$ along the spectral dimension, as shown in the following equation:

$$\mathbf{P}_i^{\text{fuse}} = \text{Concat}\left(\mathbf{P}_i, \hat{\mathbf{P}}_i\right) \in \mathbb{R}^{P \times P \times 2C}. \tag{10}$$

Accordingly, the input to the subsequent TransformerNet is a fused tensor of size $B \times P \times P \times 2C$, where the first C channels preserve the original spatial–spectral information and the remaining C channels encode spectrally suppressed features obtained via ASSM. This design enables effective collaboration between spatial-domain details and frequency-domain robustness without introducing additional weighting coefficients while maintaining compatibility with standard Transformer-based architectures.

C. DIR Contrastive Enhancement

To further enhance the discriminability of degradation-invariant representations learned by DIR-FFE, we introduce the DIR-CE. This module employs a cross-degradation SupCon learning mechanism to explicitly align embeddings of the same object under various degradation conditions while enlarging the separation between different categories. Through this process, degradation-sensitive and degradation-invariant components are effectively disentangled, leading to a more compact intraclass distribution and clearer interclass boundaries in the embedding space.

1) Batch Embedding Generation: During training, each mini-batch consists of B clean samples and their corresponding degraded versions. For each clean patch $(x_i^{\text{clean}}, y_i)$, we extract a co-located degraded counterpart x_i^{deg} generated by the multidegradation monitor. Both are fed into the shared backbone to obtain feature representations, as shown in the following equations:

$$z_i^c = \text{Proj}\left(\text{Backbone}\left(x_i^{\text{clean}}\right)\right) \tag{11}$$

$$z_i^d = \text{Proj}\left(\text{Backbone}\left(x_i^{\text{deg}}\right)\right) \tag{12}$$

where $Proj(\cdot)$ denotes a learnable projection head that maps the backbone output into a contrastive embedding space of dimension D. We concatenate the two embeddings to form the feature set, as shown in the following equation:

$$Z = \left[z_1^c, \ldots, z_B^c, z_1^d, \ldots, z_B^d\right] \in \mathbb{R}^{2B \times D}. \tag{13}$$

Equation (13) followed by L2 normalization($\|z_i\|_2 = 1$). The label set is duplicated accordingly as shown in the following equation:

$$Y = \left[y_1, \ldots, y_B, y_1, \ldots, y_B\right]. \tag{14}$$

2) Positive and Negative Pair Definition: For each anchor embedding $z_i \in Z$, we construct positive and negative sample sets under the SupCon paradigm. A positive sample is any embedding in Z sharing the same label $y_j = y_i$ but not identical to the anchor (including both clean and degraded views). A negative sample is any embedding with a different class label $y_j \neq y_i$. Formally, the positive set is defined as shown in the following equation:

$$P(i) = \left\{z_j \in Z \mid y_j = y_i, j \neq i\right\}. \tag{15}$$

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This formulation ensures that the model aligns representations across clean–degraded pairs within the same class while explicitly enforcing interclass separation.

3) SupCon Loss: For each anchor embedding z_i, the pairwise similarity between embeddings is computed using a temperature-scaled cosine similarity, as shown in the following equation:

$$s_{ij} = \frac{z_i^\top z_j}{\tau} \tag{16}$$

where τ denotes the temperature hyperparameter controlling distribution sharpness.

The per-sample SupCon loss is then expressed as shown in the following equation:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{2B}\sum_{i=1}^{2B}\frac{1}{|P(i)|}\sum_{p\in P(i)}\log\frac{\exp\left(s_{ip}\right)}{\sum_{a\in Z, a\neq i}\exp\left(s_{ia}\right)} \tag{17}$$

where the denominator sums over all samples in Z except the anchor itself. This objective encourages embeddings from the same class (clean and degraded) to cluster tightly while pushing apart features of different classes, thereby enhancing class-aware alignment and degradation invariance.

4) Training Objective: In addition to contrastive learning, the TransformerNet backbone is equipped with a classification head for supervised HSI classification. Given the fused feature representation of each sample, the classification head produces logits of dimension K, where K denotes the number of land-cover classes. For a mini-batch of size B, the classifier output is therefore of size $B \times K$. The classification loss is defined using the standard cross-entropy (CE) loss, as shown in the following equation:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B}\sum_{i=1}^{B}\log p_{i,y_i} \tag{18}$$

where p_{i,y_i} denotes the predicted probability of sample i corresponding to its ground-truth class $y_i \in \{1, \ldots, K\}$.

The final training objective is formulated as shown in the following equation:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{sup}}\mathcal{L}_{\text{sup}} \tag{19}$$

where λ_{sup} balances classification supervision and cross-degradation contrastive alignment.

III. Experiments

A. Experimental Settings

1) Datasets: In order to verify the effectiveness of our algorithm, we applied the algorithm to three public datasets: the Pavia University (PU) dataset, the WHU-Hi-LongKou (WHLK) dataset, and the Indian Pines (IP) dataset.

The PU dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the urban area of Pavia, northern Italy. The scene was captured with a spatial resolution of 1.3 m and consists of 610 $\times$ 340 pixels. It originally contains 115 spectral bands ranging from 430 to 860 nm, among which 12 bands affected by serious noise or water absorption were removed, leaving 103 effective bands for analysis. The dataset comprises nine land-cover categories, including asphalt, meadows, trees, self-blocking bricks, bare soil, bitumen, tiles, shadows, and metal sheets. Due to its diverse surface materials and fine-grained spatial structures, the PU dataset has become one of the most representative benchmarks for evaluating HSI classification and domain generalization methods. The false-color image and the classification ground truth are shown in Fig. 2.

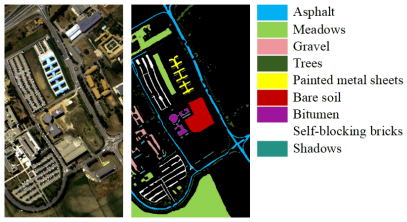


Fig. 2. Visualization of the PU dataset: false-color composite image, ground-truth map, and class legend.

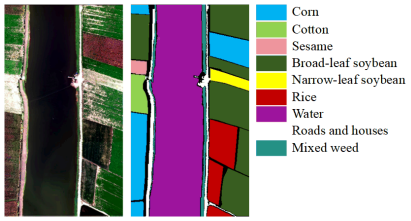


Fig. 3. Visualization of the WHLK dataset: false-color composite image, ground-truth map, and class legend.

The WHLK dataset [57], [58] was collected between 13:49 and 14:37 on July 17, 2018, over Longkou Town in Hubei Province, China. The hyperspectral imagery was captured using a Headwall Nano-Hyperspec sensor equipped with an 8-mm focal length lens, mounted on a DJI Matrice 600 Pro uncrewed aerial vehicle (UAV) flying at an altitude of approximately 500 m. The dataset consists of images with a spatial size of 550 $\times$ 400 pixels and 270 spectral bands covering the wavelength range from 400 to 1000 nm, yielding a ground sampling distance of about 0.463 m. The captured scene represents a complex agricultural landscape characterized by heterogeneous spatial textures and mixed crop distributions. It includes nine land-cover categories such as corn, cotton, sesame, roads, trees, and bare soil. Owing to its high spatial resolution and diverse spectral variability, the WHLK dataset serves as a challenging benchmark for evaluating HSI classification and robustness under real-world degradation conditions. The false-color image and the classification ground truth are shown in Fig. 3.

The IP dataset was acquired in 1992 by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over an agricultural area in northwestern Indiana, USA. The original HSI consists of 224 spectral bands covering wavelengths from 0.4 to 2.5 μm, with a spatial size of 145 $\times$ 145 pixels and a ground sampling distance of 20 m. The scene includes 16 distinct land-cover categories. Following common practice, bands severely affected by water absorption and sensor noise were discarded, resulting in a reduced dataset with 200 effective spectral bands used in our experiments. The false-color image and the classification ground truth are shown in Fig. 4.

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Fig. 4. Visualization of the IP dataset: false-color composite image, ground-truth map, and class legend.

2) Degradation Scenarios: To simulate various HSI degradation phenomena commonly encountered in real-world scenarios, nine types of degraded data were generated for this study using the multidegradation simulator (MDS) proposed in the HyperTTA [52] framework, as shown in Fig. 5.

1) *JPEG Compression:* This degradation, caused by lossy compression during HSI storage or transmission, leads to detail loss and blocky artifacts, especially in high-frequency regions. In practice, JPEG compression is applied in a band-wise manner, where each spectral band $X^{(c)} \in \mathbb{R}^{H \times W}$ is independently treated as a single-channel image and compressed as in the following equation:

$$\tilde{X}^{(c)} = \mathrm{JPEG}_q\left(X^{(c)}\right) \tag{20}$$

with q denoting the compression quality factor.

2) *Zero-Mean Gaussian Noise:* This noise, commonly caused by sensor thermal effects or electromagnetic interference, affects all spectral bands. It is simulated by adding band-wise zero-mean Gaussian noise to each spectral band $X^{(c)}$ as in the following equation:

$$\tilde{X}^{(c)} = X^{(c)} + \epsilon, \quad \epsilon \sim \mathcal{N}\left(0, \sigma^2\right) \tag{21}$$

where σ controls the overall noise intensity.

3) *Additive Gaussian Noise:* This common Gaussian noise, independent of the original signal, is simulated by adding zero-mean noise with adjustable variance to each normalized spectral band, where a larger standard deviation σ produces stronger degradation.

4) *Poisson Noise:* It originates from the quantum randomness of photon detection and is commonly observed in HSI under low-light or nighttime conditions. It is simulated by applying a Poisson process to each spectral band $X^{(c)}$, where the noisy observation is generated as in the following equation:

$$\tilde{X}^{(c)} \sim \mathrm{Poisson}\left(\lambda^{(c)}\right) \tag{22}$$

with $\lambda^{(c)}$ denoting the intensity parameter proportional to the original signal. The noise level is controlled by a target signal-to-noise ratio (SNR), where lower SNR values correspond to more severe degradation.

5) *Salt-and-Pepper Noise:* This impulse-type degradation is often caused by dead pixels, transmission errors, or scanner malfunctions and is characterized by randomly distributed extremely bright and dark pixels. It is simulated by randomly replacing a proportion p of pixels in each spectral band $X^{(c)}$ with minimum or maximum intensity values, while the remaining pixels remain unchanged. A larger value of p results in more severe impulse corruption.

6) *Stripe Noise:* It is a common imaging degradation caused by column sensor failures, transmission errors, or scanning artifacts and typically appears as vertical stripe patterns. It is simulated by randomly selecting a number of columns in each spectral band $X^{(c)}$, where the number of stripe columns n is sampled from a uniform distribution $n \sim \mathcal{U}(a, b)$. The selected columns are replaced with fixed intensity values, as shown in the following equation:

$$\tilde{X}^{(c)}_{:,j} = \lambda, \quad j \in \mathcal{J} \tag{23}$$

where $\mathcal{J}$ denotes the set of stripe columns and λ is a constant intensity. Larger values of n result in more severe stripe artifacts.

7) *Deadline Noise:* It is a stripe-like degradation caused by sensor malfunction or data transmission errors and is characterized by the complete loss of pixel values in certain image columns. It is simulated by randomly selecting a number of column positions in each spectral band $X^{(c)}$, where the number of affected columns n is sampled from a uniform distribution $n \sim \mathcal{U}(a, b)$. For each selected position j, a contiguous set of columns is set to zero as shown in the following equation:

$$\tilde{X}^{(c)}_{:,j:j+w-1} = 0 \tag{24}$$

where w denotes the width of the dead column region. Larger values of n or w lead to more severe deadline degradation.

8) *Convolutional Blur:* This type of degradation typically arises from low-pass filtering during the sensor imaging process, lens defocus, or information loss during data transmission. It is simulated by convolving each spectral band $X^{(c)}$ with a spatial averaging kernel as shown in the following equation:

$$\tilde{X}^{(c)} = X^{(c)} * K \tag{25}$$

where K denotes a $k \times k$ mean filter. The kernel size k controls the blur strength, with larger values producing stronger smoothing and more severe loss of high-frequency spatial details.

9) *Fog Degradation:* This degradation primarily results from the scattering and absorption of light by water vapor and suspended particles in the atmosphere. It is simulated using an atmospheric scattering model, where each spectral band $X^{(c)}$ is degraded as shown in the following equation:

$$\tilde{X}^{(c)} = X^{(c)} \cdot t^{(c)} + A^{(c)} \cdot \left(1 - t^{(c)}\right) \tag{26}$$

with $t^{(c)}$ denoting the transmission coefficient and $A^{(c)}$ the atmospheric light. The fog severity is controlled by a scattering parameter ω, where larger values correspond to denser fog and stronger degradation.

Table I summarizes the detailed parameter settings of all degradation types for the PU, WHLK, and IP datasets. Since these datasets differ substantially in spatial size, spatial resolution, spectral dimensionality, and sensor characteristics,

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TABLE I
DEGRADATION PARAMETER SETTINGS ON THE PU, WHLK, AND IP DATASETS

Degradation Type	Parameter	Meaning	PU Value	WHLK Value	IP Value
JPEG Compression	q	Quality factor	10	15	10
Zero-Mean Gaussian Noise	σ	Standard deviation of zero-mean Gaussian noise	0.3	0.3	0.3
Additive Gaussian Noise	σ	Standard deviation of additive noise	0.2	0.3	0.3
Poisson Noise	SNR	Signal-to-noise ratio (in dB)	30	9	7
Salt-and-Pepper Noise	p	Noise density (probability of corruption)	0.1	0.15	0.15
Stripe Noise	a, b	Range of stripe counts	[30,35]	[30,35]	[3,5]
Deadline Noise	a, b	Range of dead columns	[30,35]	[30,35]	[3,5]
Convolutional Blur	k	Kernel size for blurring	10	5	2
Fog Degradation	ω	Atmospheric scattering coefficient	0.5	0.5	0.5

some degradation parameters are set in a dataset-adaptive manner rather than fixed to identical numeric values across all datasets. The purpose is to maintain roughly comparable effective corruption severity and to avoid unrealistically weak or excessively destructive perturbations on a specific dataset. Importantly, for each dataset, all compared methods are evaluated under the same degraded samples and parameter settings, ensuring a fair comparison.

3) Evaluation Metrics: We will compare the effectiveness of our algorithm with other algorithms from three aspects, mainly including overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ).

4) Implementation Details: All experiments were conducted on a workstation equipped with an Intel[1] Xeon[1] Platinum 8336C CPU @ 2.30 GHz, 64 cores, 128 GB of memory, and two NVIDIA GeForce RTX 4090 GPUs, each with 24 GB of memory. The proposed model was implemented using the Pytorch framework.

During training, the patch and batch sizes were set to 13 and 256 for the PU dataset and IP dataset, and 33 and 64 for the WHLK dataset, respectively, to accommodate the spatial and spectral characteristics of each dataset. The training and validation sets were randomly divided based on pixel coordinates, with the training set comprising 20% of the total samples. Z-score normalization was applied using the channel-wise mean (μ) and standard deviation (σ) computed from the training set, and the same normalization statistics were consistently used for the validation, test, and degraded datasets.

Model optimization was performed using the AdamW optimizer with a learning rate of 1×10^{-3} and a weight decay of 1×10^{-4} for 150 epochs. A five-epoch linear warm-up followed by cosine annealing was adopted for learning rate scheduling, and the gradient clipping threshold was set to 1.0. The CE loss was computed only on clean samples to serve as a stable classification anchor.

For contrastive learning, a SupCon strategy was employed. Each batch contained paired clean and degraded samples, which were jointly passed through a two-layer multi-layer perceptron (MLP) projection head (projection dimension = 128, followed by L2 normalization) to produce concatenated embeddings $Z \in \mathbb{R}^{2B \times D}$ and labels Y. The SupCon loss was computed with a temperature parameter $\tau = 0.1$. The SupCon loss weight $\lambda_{\text{sup}} = 0.1$ was linearly ramped up during the first ten epochs to prevent early dominance over the CE loss.

[1]Registered trademark.

B. Performance Analysis

To evaluate the effectiveness of the proposed method, a variety of representative algorithms were selected as comparative experiments, including HyperTTA [52], SSEFN [59], SSFTT [60], SLCGC [61], RPCL-FSL [62], SS-MTr [63], and DSFormer [64]. All models were trained under identical conditions on clean data and evaluated across various simulated degradation scenarios on the PU, WHLK, and IP datasets. This comparative analysis aims to demonstrate the inherent robustness of DIR-HSI in challenging environments, thereby validating the effectiveness of the proposed DIR framework in accurately extracting degradation-invariant representations.

1) HyperTTA [52] is a unified and robust classification framework designed for HSI degradation scenarios, aiming to enhance model adaptability to various degradations such as noise, blur, compression, and atmospheric effects. The method introduces a lightweight test-time adaptation module, CELA, which enables rapid adaptation without access to source data. In this work, we adopt the same degradation benchmark as HyperTTA to facilitate a fair comparison of model robustness.
2) SSEFN [59] is a spatial–spectral enhancement and fusion network designed for HSI classification under limited labeled samples. It enhances feature discriminability by jointly exploiting multiscale spatial context and spectral differential information, and employs an adaptive decision fusion strategy to alleviate overfitting caused by data scarcity.
3) SSFTT [60] effectively combines CNN and transformer and introduces a Gaussian weighted feature tokenizer to further enhance the results. It is one of the transformer-based algorithms with Competitive performance in the current HSI classification task.
4) SLCGC [61] is a self-supervised graph clustering framework for HSI that achieves efficient and robust clustering without labeled data. It aggregates pixels into spectrally homogeneous regions to preserve spatial–spectral consistency while reducing graph complexity.
5) RPCL-FSL [62] is a refined prototypical contrastive learning framework designed for few-shot HSI classification. It integrates SupCon learning with prototypical

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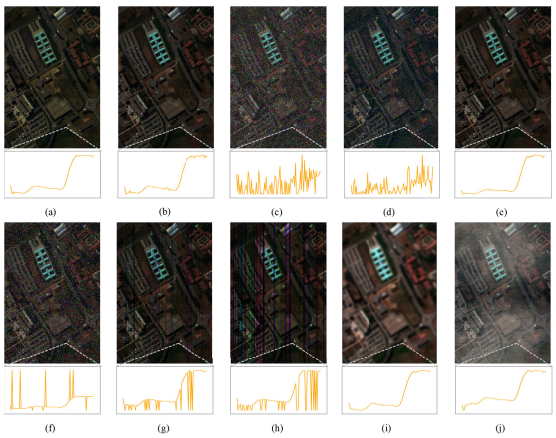


Fig. 5. Visualization of HSI with multiple degradations on PU datasets. (a) Clean, (b) JPEG, (c) zero-mean Gaussian, (d) additive Gaussian, (e) Poisson, (f) salt and pepper, (g) stripes, (h) deadline, (i) convolutional blur, and (j) fog. The spectral reflectance curves shown below each image correspond to a representative pixel located at (row = 480 and column = 220).

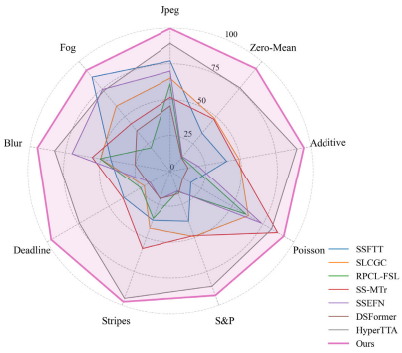


Fig. 6. Radar chart of classification performance (OA) on the PU dataset under nine types of degradation.

networks in an end-to-end manner to improve prototype stability and alleviate domain shift.

6) SS-MTr [63] is a Transformer-based HSI classification model that adopts a two-stage training strategy. It first pretrains a Transformer through masked spectral–spatial reconstruction to embed local inductive bias, and then incorporates contrastive and supervised objectives to enhance feature discriminability.
7) DSFormer [64] is a transformer-based HSI classification framework that aims to address the limitations of fixed receptive fields and redundant self-attention in complex HSI scenes. It introduces a dual selective fusion mechanism, including kernel selective fusion and token selective fusion, to adaptively capture informative spatial–spectral contexts while suppressing irrelevant or redundant features.

1) Quantitative Results: Table II presents the classification performance (OA, AA, and κ) of different methods under multiple degradation conditions on the PU dataset, and the visualized OA results are shown in Fig. 6. Compared with all baseline methods, the proposed DIR-HSI achieves consistently superior performance across most degradation types.

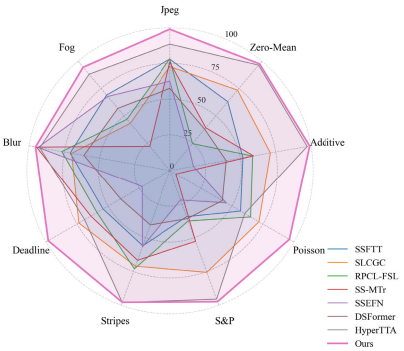


Fig. 7. Radar chart of classification performance (OA) on the WHLK dataset under nine types of degradation.

Specifically, it outperforms all competitors under challenging conditions such as additive noise (94.97%), Poisson noise (91.76%), and deadline artifacts (96.54%) while maintaining near-clean performance under milder degradations such as JPEG compression (99.56%) and stripe noise (97.37%). Overall, DIR-HSI attains the highest AA, with OA = 94.76%, AA = 91.79%, and $\kappa \times 100 = 93.34$, demonstrating its strong robustness and the effectiveness of the proposed DIR mechanism.

In comparison, HyperTTA achieves average OA, AA, and $\kappa \times 100$ values of 83.06%, 73.08%, and 77.12, respectively, performing well in certain scenarios but still falling short of DIR-HSI overall. Other baseline models, including SS-MTr, RPCL-FSL, SLCGC, SSFTT, SSEFN, and DSFormer, exhibit considerable performance fluctuations, particularly under severe degradations such as zero-mean and salt noise. These results indicate that DIR-HSI maintains stable classification performance across diverse degradation types, highlighting its superior cross-degradation generalization and robustness.

Table III summarizes the classification results of all competing methods under various simulated degradations on the WHLK dataset, while the visualized OA performance is illustrated in Fig. 7. The proposed DIR-HSI consistently achieves the best performance across nearly all degradation scenarios. Specifically, it attains superior accuracy under challenging conditions such as Poisson noise (96.25%), fog (94.39%), and blur (95.29%), while maintaining almost clean-level accuracy under mild degradations, including JPEG compression (98.83%) and additive noise (98.99%). Overall, DIR-HSI obtains the highest mean accuracy with OA = 97.15%, AA = 90.53%, and $\kappa \times 100 = 96.40$, demonstrating strong robustness and stable generalization to diverse degradation patterns.

Table IV reports the quantitative classification results of all competing methods on the IP dataset under nine representative degradation scenarios, and the corresponding OA radar visualization is shown in Fig. 8. As can be observed, the proposed DIR-HSI consistently achieves the best performance across almost all degradation types, significantly outperforming all baseline methods by a large margin. In particular, DIR-HSI maintains near-clean accuracy under mild degradations such as JPEG compression (98.85%), additive noise (97.95%), stripe

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TABLE II

COMPARISON OF CLASSIFICATION PERFORMANCE ON THE PU DATASET UNDER DIFFERENT DEGRADATION TYPES

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog	Avg.
HyperTTA	OA (%)	89.19±2.30	75.98±2.40	90.03±2.06	81.96±2.22	85.93±2.27	94.82±2.21	73.84±2.40	82.50±2.19	73.28±2.32	83.06±2.12
	AA (%)	80.16±4.45	60.16±2.43	83.96±2.56	66.18±2.30	76.04±2.31	89.46±2.40	70.60±2.36	76.75±2.37	54.35±2.35	73.08±2.20
	Kappa×100	84.97±2.33	67.64±1.87	85.52±2.31	76.03±2.09	80.75±2.27	93.61±2.03	66.90±1.70	75.52±2.15	63.11±2.06	77.12±1.81
SSEFN	OA (%)	69.87±3.22	13.31±2.97	20.05±3.23	73.81±2.64	15.14±2.33	21.38±2.76	16.53±2.32	70.15±2.85	74.50±2.58	41.64±2.20
	AA (%)	75.92±2.38	12.84±3.82	19.94±3.78	86.88±3.34	14.42±3.79	22.54±3.50	16.97±3.49	81.09±3.52	81.93±3.54	45.83±2.95
	Kappa×100	63.54±2.15	3.40±2.53	11.92±2.55	68.73±3.82	5.29±2.47	11.92±3.45	7.80±2.55	64.49±3.49	69.39±3.61	34.05±2.41
SSFTT	OA (%)	76.95±1.20	34.83±1.16	40.23±1.44	16.68±0.55	37.83±0.57	37.04±0.68	37.85±1.22	42.42±0.65	85.77±1.20	45.51±0.88
	AA (%)	66.63±1.15	49.62±0.85	53.40±1.10	16.98±0.97	47.68±1.18	43.05±0.59	34.88±1.18	29.55±0.94	78.68±0.62	46.72±0.89
	Kappa×100	69.71±1.14	27.54±1.11	32.57±0.71	10.62±1.07	30.11±0.77	26.83±0.83	24.83±1.11	19.50±1.06	81.71±1.10	35.94±0.95
SLCGC	OA (%)	64.95±2.63	48.28±1.72	48.95±2.19	63.25±3.04	49.06±2.12	42.93±1.52	21.46±2.74	49.89±2.02	59.36±1.76	49.79±1.69
	AA (%)	59.38±3.36	49.97±1.25	52.75±2.41	58.43±3.27	53.15±2.40	49.61±3.64	28.29±3.59	43.12±1.33	59.51±3.41	50.47±1.90
	Kappa×100	55.53±3.90	38.36±2.51	39.48±2.68	54.04±3.77	39.85±2.45	33.18±2.59	11.90±2.45	38.60±2.52	51.47±2.52	40.27±2.20
RPCL-FSL	OA (%)	60.93±2.24	12.77±1.81	16.12±3.49	61.35±1.90	15.61±2.23	35.73±2.56	24.26±2.00	50.41±3.03	21.45±2.56	33.18±2.15
	AA (%)	61.24±2.15	14.45±3.52	27.93±2.25	62.03±2.51	24.03±2.14	46.94±2.58	34.69±3.57	49.71±3.20	22.42±1.42	38.16±2.34
	Kappa×100	51.11±2.60	4.24±2.69	7.95±2.40	51.45±2.69	7.12±2.73	25.42±2.53	15.55±2.47	38.60±2.75	9.07±1.32	23.39±2.43
SS-MTr	OA (%)	51.56±2.80	47.57±3.88	46.62±3.55	86.86±3.73	48.42±3.16	57.98±3.60	38.63±2.87	55.79±4.63	43.27±3.16	52.97±2.80
	AA (%)	23.53±1.73	22.83±3.08	19.57±1.92	84.27±2.88	22.66±4.68	61.41±5.33	15.56±3.75	44.11±4.78	31.86±3.50	36.20±2.56
	Kappa×100	33.47±4.11	9.35±3.05	23.72±3.20	86.71±2.42	11.66±3.83	45.50±2.89	10.86±3.52	39.87±3.91	30.45±3.36	32.40±2.62
DSFormer	OA (%)	45.62±1.16	12.38±2.33	12.74±1.06	10.24±0.92	17.25±2.61	20.93±2.26	18.01±1.38	25.34±0.99	37.02±0.60	22.17±0.87
	AA (%)	55.00±1.46	16.52±1.89	26.91±2.47	18.43±0.73	26.27±2.73	38.69±0.88	33.03±1.41	40.01±1.40	43.07±1.43	33.10±0.49
	Kappa×100	36.94±1.76	7.63±1.45	7.85±1.43	5.28±1.54	13.29±1.59	14.41±1.09	11.41±1.53	18.87±0.75	28.32±1.07	16.00±0.54
Ours	OA (%)	**99.56±0.15**	**93.41±1.63**	**94.97±1.62**	**91.76±0.59**	**92.67±1.10**	**97.37±1.04**	**96.54±1.57**	**94.65±0.26**	**91.90±2.12**	**94.76±0.82**
	AA (%)	**98.99±0.39**	**86.57±2.43**	**93.20±1.01**	**87.38±1.45**	**87.67±1.16**	**96.33±1.32**	**95.67±1.68**	**90.30±0.88**	**90.00±0.58**	**91.79±0.47**
	Kappa×100	**99.42±0.19**	**90.79±1.72**	**94.31±0.90**	**89.73±0.54**	**90.13±1.51**	**97.12±1.36**	**95.91±1.81**	**91.78±1.49**	**90.84±1.55**	**93.34±0.60**

TABLE III

COMPARISON OF CLASSIFICATION PERFORMANCE ON THE WHLK DATASET UNDER DIFFERENT DEGRADATION TYPES

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog	Avg.
HyperTTA	OA (%)	88.46±2.81	96.28±1.62	97.25±2.18	60.36±2.71	95.44±2.63	97.84±1.27	68.92±3.73	92.30±2.29	87.96±2.74	87.20±0.61
	AA (%)	91.49±2.64	88.66±2.93	94.35±1.69	74.35±1.39	91.73±3.52	95.81±2.54	75.81±3.03	83.49±2.65	67.37±3.68	84.78±1.13
	Kappa×100	86.83±3.13	93.88±2.13	97.28±1.45	55.12±2.77	95.17±1.89	97.14±2.29	62.41±4.36	89.94±1.44	84.11±2.97	84.65±0.97
SSEFN	OA (%)	62.52±4.31	20.81±4.41	16.81±2.67	45.50±2.77	22.03±2.94	56.76±3.89	22.56±3.04	93.52±2.35	67.69±2.79	45.36±2.33
	AA (%)	57.52±4.89	18.06±6.09	18.89±4.06	34.56±4.98	17.05±3.06	43.17±1.89	18.67±2.12	90.66±2.55	63.83±3.18	40.27±1.04
	Kappa×100	56.69±3.24	11.03±3.45	5.03±3.71	34.89±4.56	8.51±3.02	49.12±3.44	12.75±3.28	91.21±3.44	61.04±2.83	36.70±2.22
SSFTT	OA (%)	77.96±0.39	62.92±2.91	51.79±2.51	56.97±2.86	34.47±1.84	55.98±1.66	54.07±2.17	70.82±1.47	69.00±1.50	59.33±1.35
	AA (%)	46.33±1.62	33.93±2.77	27.37±1.81	34.58±2.67	22.11±2.63	27.12±2.63	30.15±0.95	49.73±2.64	49.24±0.87	35.62±1.77
	Kappa×100	71.86±2.67	51.40±2.98	33.63±4.22	40.70±3.23	8.80±0.92	40.06±0.96	39.19±2.69	60.85±0.78	58.11±2.76	44.96±1.61
SLCGC	OA (%)	72.74±1.60	73.48±2.78	71.21±3.73	71.79±0.45	75.62±1.69	71.07±0.81	73.52±1.75	69.81±2.74	43.25±1.84	69.17±0.72
	AA (%)	48.09±2.65	46.04±2.91	44.10±0.95	49.54±0.99	48.36±2.51	53.47±0.79	51.81±2.60	54.68±2.56	42.18±0.88	48.70±1.45
	Kappa×100	65.30±1.74	64.62±3.17	61.49±2.85	64.09±0.44	67.78±0.93	63.44±0.42	66.08±1.40	62.15±3.65	35.23±1.27	61.13±0.78
RPCL-FSL	OA (%)	78.13±0.40	24.71±1.67	58.68±1.33	65.06±1.90	37.61±1.20	73.05±0.87	58.96±1.19	76.71±1.10	46.93±1.22	57.76±0.74
	AA (%)	70.03±1.27	20.64±1.05	46.46±0.83	59.41±1.03	28.57±0.65	65.10±1.15	55.75±1.47	66.93±0.59	36.88±0.75	49.97±0.46
	Kappa×100	72.58±0.89	16.68±1.98	49.95±1.40	57.39±2.23	28.04±1.42	66.33±1.35	50.71±1.35	70.65±1.32	36.30±1.36	49.85±0.84
SS-MTr	OA (%)	75.84±1.91	39.15±1.73	59.28±0.42	5.01±1.21	52.70±1.85	66.70±1.85	63.90±0.41	94.53±1.40	22.17±1.16	53.25±1.06
	AA (%)	77.17±1.13	29.18±0.96	54.05±1.48	11.14±1.46	37.86±1.12	75.05±1.16	67.62±1.47	80.99±0.85	15.33±1.46	49.82±0.73
	Kappa×100	70.11±2.12	29.67±2.09	51.03±0.70	-9.20±0.84	42.62±2.13	60.10±2.12	56.21±0.66	93.41±1.25	6.75±1.35	44.52±1.05
DSFormer	OA (%)	57.53±1.03	37.89±0.40	39.96±1.78	42.33±1.81	35.91±1.18	40.53±0.72	39.90±1.20	61.11±1.04	56.84±0.42	45.78±0.58
	AA (%)	44.09±0.53	25.18±1.46	25.06±1.70	25.31±1.39	25.21±0.69	28.71±1.05	29.95±1.61	52.97±0.54	46.21±0.56	33.63±0.69
	Kappa×100	47.23±1.27	27.50±0.65	27.89±2.07	30.58±2.12	25.90±1.38	29.72±0.67	29.30±1.37	52.24±1.27	46.91±0.66	35.25±0.60
Ours	OA (%)	**98.83±0.58**	**97.39±0.92**	**98.99±0.34**	**96.25±0.88**	**97.30±0.75**	**97.74±0.54**	**98.14±0.84**	**95.29±0.87**	**94.39±0.95**	**97.15±0.65**
	AA (%)	**96.06±0.49**	**91.30±0.93**	**97.30±0.80**	**83.19±0.98**	**94.99±0.74**	**96.56±0.51**	**93.80±0.88**	**86.65±0.89**	**74.94±0.61**	**90.53±0.68**
	Kappa×100	**98.60±0.64**	**96.66±1.03**	**98.73±0.57**	**94.66±0.41**	**96.47±0.71**	**98.67±0.59**	**96.62±1.09**	**94.65±0.97**	**92.54±0.60**	**96.40±0.65**

noise (97.67%), and blur (98.34%), demonstrating strong robustness to frequency-related and structural distortions.

More importantly, under severe degradation conditions where most baseline methods suffer dramatic performance drops, DIR-HSI still preserves stable and competitive accuracy. For example, it achieves 73.80% OA under Poisson noise and 96.46% OA under deadline artifacts, whereas most competing methods fail to generalize effectively in these challenging scenarios. Overall, DIR-HSI attains the highest average performance on the IP dataset, with OA = 95.13%, AA = 88.82%, and $\kappa \times 100 = 93.78$, clearly validating its superior degradation robustness and cross-degradation generalization capability.

In contrast, existing methods such as HyperTTA, SS-MTr, and SLCGC exhibit strong sensitivity to specific degradation types, resulting in large performance fluctuations across different scenarios. Other baselines, including SSEFN, SSFTT, RPCL-FSL, and DSFormer, show particularly poor robustness under noise-intensive degradations such as zero-mean Gaussian, Poisson, and salt-and-pepper noise. These results further confirm that DIR-HSI effectively mitigates degradation-induced domain shifts and maintains consistent discriminative representations, even on the challenging IP dataset with complex spectral variability and limited spatial resolution.

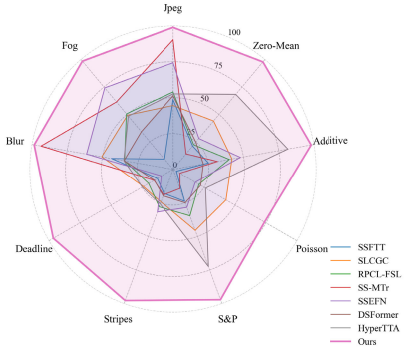


Fig. 8. Radar chart of classification performance (OA) on the IP dataset under nine types of degradation.

2) Visual Evaluation: The visual classification maps of the PU, WHLK, and IP datasets under different degradation types are shown in Figs. 9–11, respectively. To facilitate a

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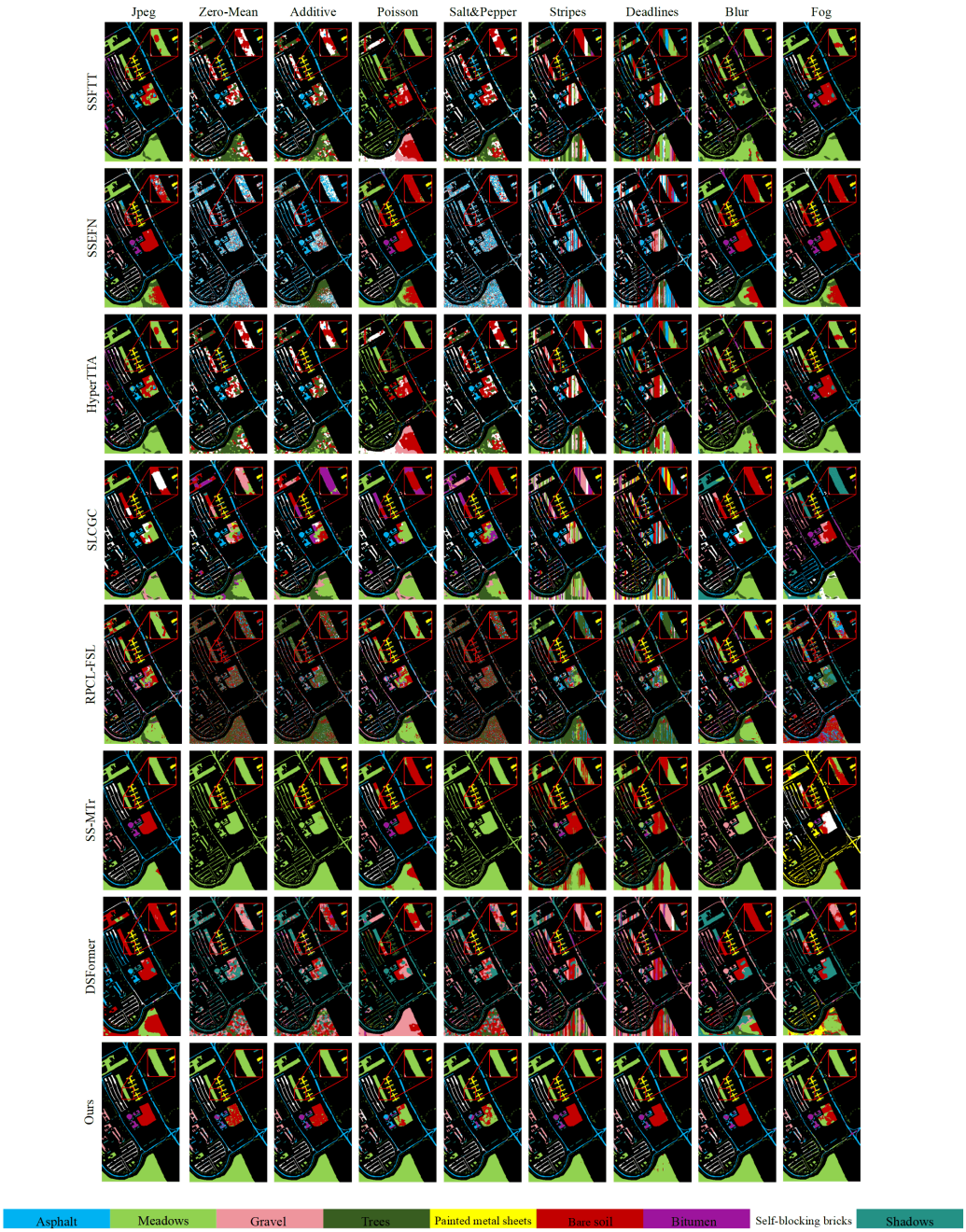


Fig. 9. Visual classification results of the PU dataset under different degradation types using various model algorithms. Each row corresponds to a specific method, and each column represents a particular type of degradation.

clearer inspection of local structural details and classification boundaries, representative regions of interest are selected and displayed in a magnified manner for all three datasets. It can be clearly observed that the proposed DIR-HSI yields the cleanest and most spatially coherent classification results across all degradation conditions. In contrast, the baseline methods produce noticeable noise, class confusion, and boundary distortions, particularly under severe degradations such as additive noise, Poisson noise, and stripe corruption. DIR-HSI effectively preserves the spatial integrity of homogeneous regions and maintains clear class boundaries even in highly degraded environments. These visual results further verify the model's strong robustness and its ability to extract degradation-invariant features, leading to visually consistent and semantically accurate classification maps on three datasets.

Fig. 12 presents the t-SNE-based feature visualization of the PU dataset under different degradation conditions. Subfigures (a)–(e) correspond to JPEG compression, deadline, Poisson, fog, and zero-mean degradations, respectively. The first row illustrates the initial degraded feature distributions, while the second row shows the classification features extracted by the proposed DIR-HSI model.

As shown in Fig. 12, the initial degraded features exhibit significant interclass overlap and blurred boundaries under

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Fig. 10. Visual classification results of the WHLK dataset under different degradation types using various model algorithms. Each row corresponds to a specific method, and each column represents a particular type of degradation.

all degradation types, indicating that the degradation noise heavily distorts the feature space. In contrast, after processing with DIR-HSI, the feature distributions become much more compact and well-separated, with clearer class boundaries and improved intraclass consistency. Notably, even under challenging degradations such as Poisson noise and fog, the model maintains distinct and robust feature clustering. These results demonstrate the effectiveness of the proposed DIR-FFE and DIR-CE in suppressing degradation-induced distortions, enhancing feature discriminability, and preserving cross-degradation consistency.

C. Ablation Study

Tables V–VII present the ablation study results on the two core components of the proposed DIR-HSI classification framework. To evaluate their individual contributions, we removed each module independently and jointly and then assessed the model performance under various degradation types on three datasets.

As shown in Table V, the proposed DIR-HSI consistently achieves the highest OA across almost all degradation types on the PU dataset. The complete model maintains stable performance above 90% OA even under severe degradations such as additive noise, Poisson noise, and fog, demonstrating strong robustness and generalization ability. When examining the effect of each component, removing the DIR-FFE leads to a noticeable decline, particularly under degradations that strongly affect spectral integrity, such as zero-mean noise (OA drops from 94.35% to 85.71%) and Poisson noise (from 91.18% to 77.91%). This confirms that DIR-FFE effectively

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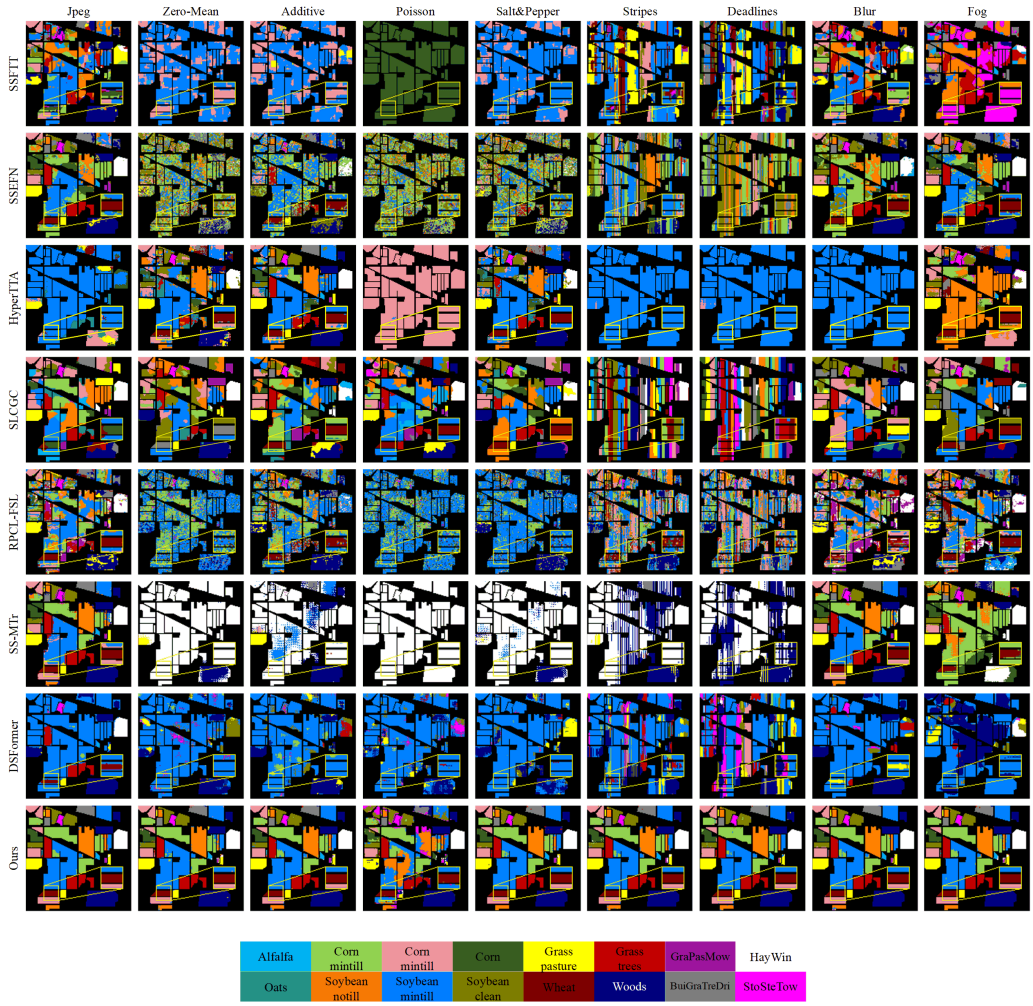


Fig. 11. Visual classification results of the IP dataset under different degradation types using various model algorithms. Each row corresponds to a specific method, and each column represents a particular type of degradation.

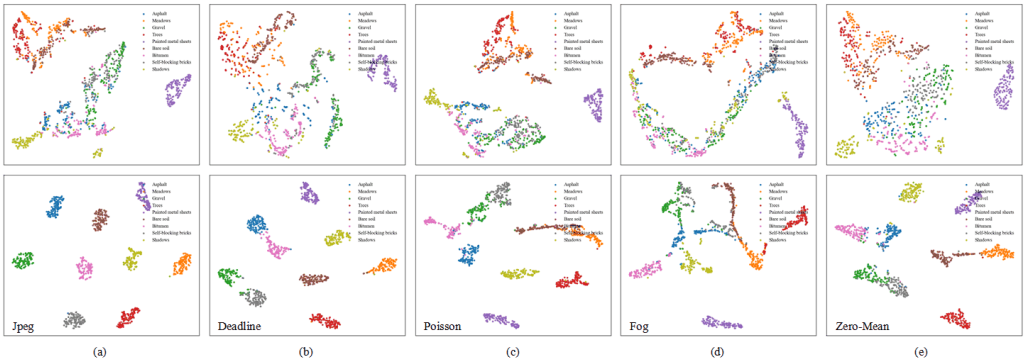


Fig. 12. Feature visualization of the PU degraded dataset based on t-SNE. (a) JPEG compression. (b) Deadlines. (c) Poisson noise. (d) Fog degradation. (e) Zeromean noise. The top-right legend lists the PU dataset's land-cover classes.

extracts degradation-invariant frequency features and enhances the model's resilience to spectral distortions. Similarly, excluding the DIR-CE results in a marked reduction in performance, especially under severe noise and low-contrast conditions like zero-mean and fog degradations (OA decreases from 94.35% to 66.93% and from 93.61% to 66.15%, respectively). This highlights the critical role of contrastive regularization in improving intraclass compactness and interclass separability. Furthermore, when both DIR-FFE and DIR-CE are removed, the performance deteriorates sharply across all degradations (average OA drops from 95.06% to 69.53%), indicating that the synergy between spectral–frequency representation and contrastive consistency is essential for achieving robust degradation-invariant learning.

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TABLE IV

COMPARISON OF CLASSIFICATION PERFORMANCE ON THE IP DATASET UNDER DIFFERENT DEGRADATION TYPES

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog	Avg.
HyperTTA	OA (%)	52.60±1.18	68.23±0.42	81.44±1.07	26.23±1.17	72.36±1.23	25.79±0.36	24.66±1.16	34.84±1.74	48.47±0.72	48.29±0.71
	AA (%)	49.92±1.45	69.96±1.47	88.49±0.57	25.59±1.46	72.88±1.34	24.36±0.44	7.32±1.49	34.05±1.27	47.01±1.05	46.62±0.78
	Kappa×100	45.19±1.44	63.96±0.68	78.61±1.28	16.01±1.38	68.11±1.40	15.55±0.68	18.52±1.40	25.79±2.08	40.45±0.35	41.35±0.81
SSEFN	OA (%)	74.28±1.68	28.12±2.40	47.81±1.63	17.93±1.46	28.36±1.60	31.67±1.62	9.26±1.51	61.22±1.81	74.11±1.66	41.42±1.47
	AA (%)	85.68±1.88	20.77±2.04	48.23±1.80	12.24±2.02	26.40±1.89	21.09±1.87	8.44±2.04	79.02±1.86	85.88±1.76	43.08±1.68
	Kappa×100	71.10±1.96	19.32±2.79	39.20±1.91	7.38±1.22	19.24±1.97	21.69±1.94	1.26±1.93	55.67±3.07	71.32±1.93	34.02±1.89
SSFTT	OA (%)	48.61±1.27	21.31±0.96	25.16±1.24	2.71±0.89	23.72±1.11	17.92±0.99	12.55±1.19	43.58±1.22	9.34±1.18	22.77±0.55
	AA (%)	38.70±1.31	7.35±0.80	9.85±1.14	6.62±0.83	7.94±1.17	10.29±1.14	8.16±0.67	35.44±1.34	17.76±1.42	15.79±0.51
	Kappa×100	40.95±1.63	2.08±1.03	8.78±1.17	1.48±1.12	4.50±1.03	5.76±1.09	3.50±1.03	35.55±1.52	5.41±1.49	12.00±0.58
SLCGC	OA (%)	44.49±1.89	44.09±1.71	41.80±2.37	42.69±1.82	45.45±2.18	24.23±1.24	23.86±1.95	50.46±2.16	49.32±2.56	40.71±1.22
	AA (%)	47.06±2.04	53.71±2.83	48.99±2.35	49.08±2.33	47.72±1.33	32.34±1.96	31.32±1.75	48.50±1.73	48.59±2.11	45.26±1.01
	Kappa×100	39.16±1.76	37.64±2.14	36.01±2.05	36.52±2.32	38.21±2.09	17.56±0.70	18.01±2.20	44.68±2.20	43.52±1.95	34.59±1.20
RPCL-FSL	OA (%)	53.96±2.95	22.41±2.23	39.93±1.68	20.84±1.96	34.56±2.40	27.90±1.84	19.59±1.74	34.35±2.05	50.51±1.80	33.78±0.77
	AA (%)	66.92±2.36	11.93±2.26	33.56±1.29	9.38±1.75	16.18±2.25	28.00±2.35	14.34±1.10	38.63±1.82	62.71±2.21	31.30±0.70
	Kappa×100	48.46±2.76	5.92±1.70	29.93±1.84	2.39±1.53	19.37±2.24	18.23±2.32	7.70±1.84	26.94±2.17	43.86±2.47	22.53±0.83
SS-MTr	OA (%)	90.23±3.12	14.08±3.09	31.57±2.96	5.54±1.96	14.15±1.76	18.88±2.08	15.18±2.33	93.54±2.36	61.34±2.80	38.28±0.84
	AA (%)	93.57±3.00	13.13±3.07	22.58±2.75	7.02±2.08	10.81±2.35	15.88±2.19	11.31±2.10	94.85±2.03	79.56±1.73	38.75±0.98
	Kappa×100	90.27±3.02	9.83±1.93	25.05±3.07	2.02±0.91	8.85±1.08	12.48±2.26	8.33±2.02	92.90±2.21	57.71±2.93	34.16±0.63
DSFormer	OA (%)	51.30±3.34	24.79±2.60	21.48±2.63	22.25±2.68	24.84±1.94	19.70±2.65	14.96±2.68	34.61±3.31	34.26±2.29	27.58±1.89
	AA (%)	39.93±3.33	7.25±2.37	7.46±2.31	6.20±2.14	7.62±1.69	8.50±2.32	6.28±2.35	17.89±3.05	22.33±2.14	13.72±1.69
	Kappa×100	39.62±2.97	4.42±1.63	5.78±2.73	3.59±1.50	3.38±1.18	2.77±1.50	2.44±1.51	18.66±2.26	16.32±1.64	10.78±1.24
Ours	OA (%)	**98.85±0.29**	**97.69±0.53**	**97.95±0.72**	**73.80±3.20**	**97.05±0.46**	**97.67±0.63**	**96.46±0.93**	**98.34±0.28**	**98.38±0.38**	**95.13±0.46**
	AA (%)	**96.54±0.84**	**89.43±1.97**	**92.58±2.93**	**67.22±3.10**	**89.25±0.81**	**89.97±1.31**	**86.40±2.62**	**94.09±1.82**	**93.95±1.04**	**88.82±0.69**
	Kappa×100	**98.69±0.33**	**97.37±0.60**	**97.67±0.82**	**64.08±2.69**	**96.63±0.52**	**97.35±0.72**	**95.95±1.07**	**98.11±0.32**	**98.15±0.43**	**93.78±0.71**

TABLE V

ABLATION STUDY ON DIR-FFE AND DIR-CE MODULES ON PU DATASET

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog
Ours	OA (%)	**99.37**	**94.35**	**95.83**	**91.18**	**91.96**	**98.29**	**96.19**	**94.75**	**93.61**
	AA (%)	**99.12**	**87.07**	**92.54**	**88.75**	**87.99**	**97.55**	**96.73**	**90.83**	**90.57**
	Kappa×100	**99.57**	**92.39**	**94.35**	**89.65**	**90.51**	**98.28**	**96.71**	**93.17**	**92.88**
w/o DIR-FFE	OA (%)	98.80	85.71	91.83	77.91	87.66	91.35	80.67	90.06	74.65
	AA (%)	97.65	79.24	90.89	63.79	82.37	90.84	86.54	83.21	69.72
	Kappa×100	97.14	82.05	90.59	76.55	83.65	89.22	78.22	88.05	65.91
w/o DIR-CE	OA (%)	96.14	66.93	81.20	55.63	81.41	89.08	75.12	85.68	66.15
	AA (%)	90.65	63.63	73.35	34.23	73.61	81.28	70.33	73.70	56.02
	Kappa×100	94.85	59.99	75.81	31.57	74.94	85.52	69.54	80.65	57.51
w/o DIR-FFE and DIR-CE	OA (%)	96.10	31.03	79.72	51.00	62.76	88.31	71.51	86.05	59.25
	AA (%)	92.80	43.60	83.08	28.81	68.78	82.10	67.24	74.61	41.60
	Kappa×100	94.80	23.02	77.90	30.65	54.92	84.58	63.67	81.27	43.35

TABLE VI

ABLATION STUDY ON DIR-FFE AND DIR-CE MODULES ON THE WHLK DATASET

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog
Ours	OA (%)	**99.25**	**97.35**	**99.14**	**96.01**	**97.64**	**97.98**	**98.12**	**95.25**	**94.35**
	AA (%)	**96.28**	**91.22**	**97.66**	**82.79**	**95.32**	**96.79**	**93.72**	**86.57**	**75.21**
	Kappa×100	**99.06**	**96.60**	**98.98**	**94.63**	**96.79**	**98.93**	**96.53**	**94.59**	**92.81**
w/o DIR-FFE	OA (%)	97.15	89.11	92.70	88.54	92.43	96.54	95.03	91.82	85.93
	AA (%)	88.98	84.67	72.62	80.03	85.59	82.57	78.34	72.94	66.81
	Kappa×100	96.24	85.31	90.24	85.76	91.20	95.42	93.40	89.20	78.44
w/o DIR-CE	OA (%)	94.91	73.17	96.91	89.65	78.26	88.48	97.61	83.71	72.95
	AA (%)	94.77	51.61	88.09	78.71	54.81	83.47	95.31	76.33	56.07
	Kappa×100	95.89	64.68	95.91	79.54	71.18	88.00	91.86	81.79	65.30
w/o DIR-FFE and DIR-CE	OA (%)	76.73	71.08	66.71	54.22	62.05	78.56	86.32	74.77	52.91
	AA (%)	79.01	25.68	64.66	66.19	42.20	74.58	78.15	66.79	53.51
	Kappa×100	70.98	54.38	59.12	51.96	68.49	78.11	82.38	73.15	50.88

Similar trends are observed on the WHLK dataset. The complete DIR-HSI achieves the best overall performance (average OA = 97.23%, AA = 90.62%, and kappa = 96.55%) across all degradation types. Removing DIR-FFE results in a large drop under zero-mean noise and fog degradations (OA from 97.35% to 89.11% and from 94.35% to 85.93%, respectively), confirming that frequency-domain robustness is crucial for maintaining spectral stability. In contrast, eliminating DIR-CE leads to unstable predictions under low-visibility conditions such as fog (OA decreases to 72.95%), demonstrating that contrastive learning plays a vital role in ensuring consistent representation alignment across degraded domains. The joint removal of both modules yields the poorest results, with overall performance dropping to an average OA of 69.20%, clearly validating the necessity of both components.

On the IP dataset, similar but more pronounced degradation-sensitive behaviors are observed, as reported in Table VII. The complete DIR-HSI consistently achieves the best performance across all degradation types, maintaining high accuracy even under severe conditions such as Poisson noise (OA = 72.95%) and deadline artifacts (OA = 96.96%). When the DIR-FFE module is removed, the classification performance degrades noticeably, particularly under noise-related degradations, including zero-mean Gaussian noise (OA decreases from 97.25% to 85.11%) and Poisson noise (from 72.95% to 64.22%), indicating that frequency-domain modeling plays a crucial role in mitigating spectral distortions on the IP dataset. Likewise, excluding the DIR-CE module leads to substantial accuracy drops under additive noise (OA decreases from 98.45% to 80.83%) and Poisson noise (from 72.95%

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TABLE VII

ABLATION STUDY ON DIR-FFE AND DIR-CE MODULES ON IP DATASET

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog
Ours	OA (%)	**98.97**	**97.25**	**98.45**	**72.95**	**97.38**	**98.13**	**96.96**	**98.54**	**98.34**
	AA (%)	**97.12**	**88.94**	**90.16**	**67.88**	**89.19**	**89.79**	**88.59**	**93.99**	**92.01**
	Kappa×100	**98.82**	**96.86**	**98.23**	**62.81**	**97.12**	**96.86**	**97.67**	**98.45**	**98.11**
w/o DIR-FFE	OA (%)	90.43	85.11	95.09	64.22	89.02	93.27	91.96	95.56	92.20
	AA (%)	86.86	71.36	89.80	56.25	78.32	85.60	76.79	89.94	86.89
	Kappa×100	88.99	82.64	94.37	40.99	87.34	92.26	90.73	95.36	91.02
w/o DIR-CE	OA (%)	96.14	75.36	80.83	55.63	79.34	82.93	73.22	85.68	75.36
	AA (%)	90.65	65.21	70.02	34.24	72.55	71.51	68.56	73.70	65.21
	Kappa×100	94.85	67.71	73.66	31.57	75.23	83.72	65.70	80.65	67.71
w/o DIR-FFE and DIR-CE	OA (%)	84.40	47.45	71.76	43.77	56.02	39.96	23.46	84.99	44.87
	AA (%)	73.20	35.92	63.65	46.25	47.73	32.48	17.15	73.48	41.50
	Kappa×100	82.20	40.64	67.89	39.55	49.97	32.98	14.06	82.88	38.02

TABLE VIII

PERFORMANCE COMPARISON OF DIFFERENT KERNEL WEIGHTING SCHEMES ON THE PU DATASET

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog
Ours	OA (%)	**99.37**	**94.35**	**95.83**	**91.18**	**91.96**	**98.29**	**96.19**	**94.75**	**93.61**
	AA (%)	**99.12**	**87.07**	**92.54**	**88.75**	**87.99**	**97.55**	**96.73**	**90.83**	**90.57**
	Kappa×100	**99.57**	**92.39**	**94.35**	**89.65**	**90.51**	**98.28**	**96.71**	**93.17**	**92.88**
Only Gaussian Kernel	OA (%)	99.34	90.35	94.21	85.48	91.95	97.92	93.95	91.16	82.13
	AA (%)	98.69	75.02	87.70	83.64	85.40	93.78	91.66	77.30	71.99
	Kappa×100	99.12	87.02	92.95	75.21	90.25	97.23	94.79	88.11	88.43
Only Ideal Low-Pass Kernel	OA (%)	99.60	88.86	95.00	90.88	89.00	98.05	97.22	92.04	84.70
	AA (%)	99.06	71.84	88.65	86.05	88.30	95.95	94.16	79.51	87.03
	Kappa×100	99.47	84.99	93.67	83.36	87.32	97.42	96.32	89.36	79.56
Only Butterworth Low-Pass Kernel	OA (%)	99.27	90.43	94.94	80.61	92.90	96.37	93.99	90.64	88.48
	AA (%)	97.54	73.47	83.16	78.56	84.76	89.65	84.28	74.53	71.82
	Kappa×100	99.03	87.20	93.24	68.13	89.11	95.17	91.94	87.37	83.13
Equal-Weighted Fixed Kernel Fusion	OA (%)	99.12	89.69	94.57	87.27	91.04	96.73	96.40	90.53	89.52
	AA (%)	98.02	72.58	86.37	82.37	86.13	97.13	91.51	85.28	86.79
	Kappa×100	99.01	86.04	93.18	85.91	88.32	96.08	95.69	88.18	90.96

to 55.63%), highlighting the importance of contrastive regularization in preserving discriminative representations under degraded conditions. When both DIR-FFE and DIR-CE are jointly removed, the performance deteriorates dramatically across all degradation types, with the OA under zero-mean and Poisson noise dropping to 47.45% and 43.77%, respectively, confirming that the synergistic integration of both modules is essential for achieving robust degradation-invariant classification on the IP dataset.

Overall, these findings demonstrate that both the DIR-FFE and DIR-CE modules are indispensable to the robustness of the DIR-HSI classification framework. Their combination enables the model to effectively suppress degradation-induced variations and maintain stable classification accuracy under diverse and challenging degradation conditions.

D. Model Analysis

1) Filter Parameter Selection: As introduced in Section II-B, the proposed ASSM incorporates three representative low-pass filtering operators, including the Ideal, Gaussian, and Butterworth filters, and enables their contributions to be adaptively adjusted through learnable parameters. To further validate the effectiveness of the proposed learnable filter parameterization, we conduct a dedicated ablation study by comparing the full model with several fixed filtering strategies, including using a single low-pass filter and an equal-weighted fusion of the three filters. The quantitative results on the PU, WHLK, and IP datasets are reported in Tables VIII–X.

On the PU dataset, the proposed learnable filtering scheme consistently achieves the best performance under almost all degradation types. In particular, under frequency-sensitive degradations such as zero-mean Gaussian noise and fog, our method reaches 94.35% and 93.61% OA, respectively, clearly outperforming the best single-filter baselines by noticeable margins. Although fixed equal-weight fusion improves over individual filters, it still underperforms the learnable scheme when facing severe degradations, including Poisson noise and fog, indicating that static weighting cannot adequately adapt to diverse spectral distortions.

Similar trends can be observed on the WHLK dataset. The proposed method maintains superior robustness under complex degradations, especially those involving atmospheric effects and stochastic noise. For instance, under fog, our approach achieves 94.35% OA, whereas single fixed filters drop to 81.06%–86.88% OA. Even compared with equal-weighted fusion, the learnable strategy yields more stable improvements across most degradation types, demonstrating its stronger adaptability to high-resolution UAV hyperspectral imagery.

The IP dataset exhibits higher spectral variability and greater sensitivity to impulsive and blur-related noise. Under such challenging conditions, fixed filtering strategies demonstrate pronounced instability. For instance, their OA typically falls below 84% under salt-and-pepper and blur degradations, whereas the proposed method achieves 97.38% and 98.54% OA, respectively. Even the equal-weight fusion strategy consistently underperforms the learnable weighting scheme across multiple degradation scenarios, indicating that fixed parameter configurations struggle to maintain effective discriminative capability in the presence of complex spectral perturbations.

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TABLE IX

PERFORMANCE COMPARISON OF DIFFERENT KERNEL WEIGHTING SCHEMES ON THE WHLK DATASET

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog
Ours	OA (%)	**99.25**	**97.35**	**99.14**	**96.01**	**97.64**	**97.98**	**98.12**	**95.25**	**94.35**
	AA (%)	**96.28**	**91.22**	**97.66**	**82.79**	**95.32**	**96.79**	**93.72**	**86.57**	**75.21**
	Kappa×100	**99.06**	**96.60**	**98.98**	**94.63**	**96.79**	**98.93**	**96.53**	**94.59**	**92.81**
Only Gaussian Kernel	OA (%)	96.54	96.60	96.14	92.64	94.44	93.91	95.71	95.18	86.88
	AA (%)	83.61	86.15	86.11	84.01	78.66	92.35	93.65	88.11	74.06
	Kappa×100	95.39	95.77	93.01	90.18	91.38	93.99	96.28	92.01	83.18
Only Ideal Low-Pass Kernel	OA (%)	97.19	97.92	97.42	92.22	93.41	95.02	96.06	94.68	85.15
	AA (%)	86.39	85.78	93.61	82.97	79.36	88.74	94.12	88.88	73.23
	Kappa×100	96.28	96.22	94.82	89.67	89.89	93.67	96.67	91.39	85.86
Only Butterworth Low-Pass Kernel	OA (%)	96.43	96.97	95.41	91.37	92.30	93.12	95.03	92.77	81.06
	AA (%)	82.03	74.51	83.55	78.57	82.09	91.98	89.99	85.90	63.61
	Kappa×100	95.26	90.04	92.04	88.50	88.93	92.01	95.37	88.61	84.42
Equal-Weighted Fixed Kernel Fusion	OA (%)	97.29	95.78	97.57	94.04	95.67	96.28	98.09	93.02	85.50
	AA (%)	94.76	89.21	93.73	80.82	93.77	93.17	90.96	81.88	71.67
	Kappa×100	95.68	91.01	97.12	90.68	94.41	96.69	94.12	89.96	85.27

TABLE X

PERFORMANCE COMPARISON OF DIFFERENT KERNEL WEIGHTING SCHEMES ON THE IP DATASET

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog
Ours	OA (%)	**98.97**	**97.25**	**98.45**	**72.95**	**97.38**	**98.13**	**96.96**	**98.54**	**98.34**
	AA (%)	**97.12**	**88.94**	**90.16**	**67.88**	**89.19**	**89.79**	**88.59**	**93.99**	**92.01**
	Kappa×100	**98.82**	**96.86**	**98.23**	**62.81**	**97.12**	**96.86**	**97.67**	**98.45**	**98.11**
Only Gaussian Kernel	OA (%)	96.58	95.72	94.69	97.89	70.99	96.79	94.97	83.62	96.81
	AA (%)	86.13	83.71	88.34	88.13	62.28	85.99	80.54	69.64	88.67
	Kappa×100	96.09	92.96	93.32	97.31	52.48	96.01	93.20	81.20	95.28
Only Ideal Low-Pass Kernel	OA (%)	97.04	96.92	96.40	96.22	67.09	93.17	93.05	83.33	94.14
	AA (%)	85.53	83.76	85.76	90.68	49.43	77.16	77.51	68.34	85.15
	Kappa×100	96.63	94.30	91.30	95.69	51.91	87.61	91.61	80.88	93.33
Only Butterworth Low-Pass Kernel	OA (%)	96.73	97.64	95.66	96.07	66.26	92.43	95.89	83.16	95.58
	AA (%)	85.46	85.12	80.96	84.99	47.47	76.83	80.87	68.62	85.43
	Kappa×100	96.27	95.13	94.46	95.51	49.60	90.76	94.29	80.67	94.90
Equal-Weighted Fixed Kernel Fusion	OA (%)	97.39	93.44	96.43	68.25	95.58	95.29	96.92	97.77	95.82
	AA (%)	93.02	79.73	88.28	63.48	80.90	85.75	85.39	92.35	88.45
	Kappa×100	97.35	81.15	94.98	55.07	92.46	94.08	94.78	95.36	96.70

TABLE XI

ZERO-SHOT GENERALIZATION PERFORMANCE ON THE PU DATASET

Method	Metric	Jpeg	Zero-Mean	Additive	Poisson	Salt&Pepper	Stripes	Deadline	Blur	Fog
Kill one	OA (%)	99.68	78.65	88.70	53.64	90.93	92.21	79.21	84.13	23.25
	AA (%)	99.13	70.03	84.47	37.84	85.80	88.40	73.43	73.74	31.68
	Kappa×100	99.58	72.36	84.46	23.98	87.75	89.76	72.73	78.76	13.87
Ours	OA (%)	**99.37**	**94.35**	**95.83**	**91.18**	**91.96**	**98.29**	**96.19**	**94.75**	**93.61**
	AA (%)	**99.12**	**87.07**	**92.54**	**88.75**	**87.99**	**97.55**	**96.73**	**90.83**	**90.57**
	Kappa×100	**99.57**	**92.39**	**94.35**	**89.65**	**90.51**	**98.28**	**96.71**	**93.17**	**92.88**

Across three datasets, these ablation results consistently demonstrate that the proposed ASSM benefits substantially from learnable filter parameters and adaptive fusion weights, rather than relying on a single handcrafted operator or fixed-weight fusion strategy. This observation can be attributed to the fact that different degradation types—such as blur, noise, compression artifacts, and atmospheric scattering—affect hyperspectral signals through distinct frequency perturbation patterns, whose impacts further vary across scenes and sensors. Consequently, a fixed low-pass operator or equal-weighted fusion scheme fails to maintain an optimal tradeoff between degradation suppression and discriminative information preservation under diverse corruption conditions. In contrast, the proposed learnable scheme dynamically adjusts both spectral attenuation characteristics and spatial–frequency fusion pReferences during training, enabling the model to adapt to heterogeneous degradation distributions. As a result, ASSM yields more stable degradation-invariant representations and achieves consistently superior robustness across multiple degradation scenarios.

2) Zero-Shot Generalization: To analyze the zero-shot transferability across different degradation types, we adopt a leave-one-degradation-out ("Kill one") protocol on the PU dataset, where one degradation category is excluded during training and used only for evaluation. In this setting, the performance reflects whether a specific degradation can be implicitly handled by representations learned from other degradation types. Meanwhile, the full DIR-HSI model, trained using all nine degradations, serves as a reference upper bound.

As shown in Table XI, the zero-shot transferability varies significantly across degradation types. For degradations such as JPEG compression, stripe noise, and salt-and-pepper noise, the "Kill one" setting still preserves relatively high accuracy, indicating that these degradations share partially overlapping frequency characteristics with other corruption types and thus benefit from implicit cross-degradation transfer. In contrast, severe performance drops are observed for Poisson noise and fog degradation, where the OA decreases to 53.64% and 23.25%, respectively. This behavior suggests that these degradations exhibit more distinctive spectral perturbation patterns—such as signal-dependent noise statistics or low-frequency energy redistribution—that cannot be sufficiently captured when they are absent from training.

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TABLE XII
COMPARISON OF CLASSIFICATION PERFORMANCE ON THE PU DATASET UNDER COMPOSITE DEGRADATION TYPES

Composite Data	Metric	HyperTTA	SSEFN	SSFTT	SCLGC	RPCL-FSL	SS-MTr	DSFormer	Ours
3 Mix	OA (%)	97.61	25.42	15.83	45.58	37.25	64.06	23.54	**98.16**
	AA (%)	92.43	27.12	16.59	51.24	54.54	73.83	25.63	**94.86**
	Kappa×100	96.84	15.36	4.69	35.50	30.08	55.26	10.78	**97.56**
5 Mix	OA (%)	52.97	35.90	11.36	44.84	36.84	74.80	24.19	**94.11**
	AA (%)	31.96	38.77	12.56	52.28	53.16	85.28	29.57	**90.26**
	Kappa×100	31.48	26.55	0.61	35.45	27.84	68.92	11.32	**92.14**
7 Mix	OA (%)	51.95	54.33	50.53	41.14	65.09	56.22	28.69	**55.26**
	AA (%)	33.02	53.86	26.29	41.15	43.48	41.57	22.81	**54.26**
	Kappa×100	22.28	51.43	16.22	30.10	22.63	38.38	10.60	**57.98**
9 Mix	OA (%)	60.10	46.69	11.76	57.87	21.67	52.18	32.03	**74.74**
	AA (%)	41.02	48.88	12.31	58.08	34.29	49.17	25.02	**60.98**
	Kappa×100	45.09	37.83	1.98	49.85	12.56	42.08	10.92	**66.60**

By comparison, the full DIR-HSI model consistently achieves strong performance across all degradation types, including Poisson noise and fog, demonstrating that explicit exposure to diverse degradation mechanisms during training enables the model to learn more comprehensive degradation-invariant representations. Overall, this zero-shot analysis highlights both the limits of cross-degradation transferability and the importance of incorporating diverse degradation patterns during training to achieve robust real-world performance.

3) Composite Degradation Evaluation: In real-world hyperspectral imaging scenarios, degradations rarely occur in isolation but are often coupled and superimposed due to sensor imperfections, atmospheric effects, and complex acquisition conditions. To further evaluate the robustness and generalization ability of the proposed DIR-HSI under such realistic settings, we construct a series of composite degradation benchmarks by mixing the nine degradation types used in this work with different compositions and weights. Specifically, four composite scenarios are considered, including thre-, five-, seven-, and nine-mix degradations, where each sample is corrupted by multiple degradation sources with randomly sampled mixing weights.

Table XII reports the classification performance on the PU dataset under these composite degradation settings. As the number of mixed degradation types increases, all compared methods experience a clear performance drop, indicating the substantial challenge posed by compound degradations. Nevertheless, DIR-HSI consistently achieves the best performance across all composite scenarios in terms of OA, AA, and κ, demonstrating strong robustness to complex degradation interactions. In the three-mix setting, DIR-HSI attains an OA of 98.16%, already outperforming the strongest competing method by a clear margin. More importantly, as the degradation complexity increases, the performance gap becomes more pronounced. Under the five- and seven-mix scenarios, most baseline methods suffer from severe instability and accuracy degradation, while DIR-HSI maintains relatively high and stable performance, achieving 94.11% and 55.26% OA, respectively. In the most challenging nine-mix scenario, which involves all nine degradation types simultaneously, DIR-HSI still achieves an OA of 74.74%, significantly surpassing all competing methods.

These results indicate that existing approaches, including denoising-based, attention-based, and test-time adaptation methods, struggle to maintain consistent representations when multiple degradation factors interact. In contrast, the proposed DIR-HSI effectively captures degradation-invariant representations by explicitly modeling frequency-domain stability and adaptive feature fusion, enabling robust classification even under highly complex and compound degradation conditions.

4) Per-Class Classification Performance: To further investigate how different degradation types influence classification performance across land-cover categories, Table XIII reports the per-class classification accuracies of the compared methods on the PU dataset under various degradation conditions.

Overall, the proposed method consistently achieves higher and more stable classification accuracy across most categories, demonstrating strong robustness to diverse degradation types. Compared with the baseline methods, our approach maintains reliable performance, particularly for classes with distinctive spectral signatures such as asphalt, meadows, trees, and metal, even under severe degradations, including stripes, deadline, and salt-and-pepper noise. In contrast, the competing methods exhibit noticeable performance degradation in several categories when the spectral structure is heavily distorted. For instance, DSFormer and SSFTT show significant accuracy drops for classes such as meadows and bitumen under strong noise conditions, indicating limited robustness to spectral perturbations.

These results demonstrate that the proposed method not only improves overall classification accuracy but also enhances the robustness of category-level recognition, enabling more reliable classification of diverse land-cover types under complex degradation conditions.

5) Parameter Sensitivity Analysis: To evaluate the influence of key hyperparameters on the proposed framework, we conduct parameter sensitivity experiments on the SupCon loss weight λ_{sup} and the temperature parameter τ used in the contrastive learning objective.

As shown in Fig. 13, when varying the contrastive loss weight λ_{sup} within the range $[0.05, 0.20]$, the classification accuracy remains relatively stable across different degradation types. However, excessively small values weaken the effect of contrastive supervision, while overly large values may introduce training instability due to excessive emphasis on the contrastive objective. Fig. 14 further analyzes the influence of the temperature parameter τ. Similar to the behavior observed for λ_{sup}, the model performance shows moderate sensitivity

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TABLE XIII
PER-CLASS CLASSIFICATION ACCURACY (%) ON THE PU DATASET UNDER DIFFERENT DEGRADATION TYPES

Method	Degradation	1:Asphalt	2:Meadows	3:Gravel	4:Trees	5:Metal	6:Soil	7:Bitumen	8:Bricks	9:Shadows	OA(%)	AA(%)	$\kappa\times100$
DSFormer	Jpeg	51.88	32.81	59.07	57.54	60.27	54.42	60.29	56.56	60.90	45.74	54.86	36.16
	Zero-Mean	22.31	0.00	16.45	33.44	8.80	27.31	6.85	31.51	5.04	13.18	16.86	8.62
	additive	24.03	0.00	11.82	19.47	43.65	24.69	41.68	17.97	38.62	13.67	24.66	8.86
	Poisson	15.88	0.00	22.51	21.10	23.62	18.23	23.64	20.20	24.20	10.97	18.82	5.37
	Salt&Pepper	28.95	0.00	45.76	42.18	19.56	34.90	28.61	39.89	15.03	19.12	28.32	13.59
	Stripes	32.21	0.00	47.50	44.25	50.05	37.62	50.10	42.16	51.39	22.80	39.48	14.13
	Deadline	27.92	0.00	39.67	37.17	41.63	32.07	41.67	35.57	35.66	19.15	32.37	11.45
	Blur	35.06	4.66	46.52	44.08	48.43	39.11	48.47	42.52	49.44	25.28	39.81	18.46
	Fog	40.84	27.81	45.75	44.70	46.56	42.57	46.58	44.03	46.99	36.64	42.87	27.96
HyperTTA	Jpeg	92.83	95.83	82.83	92.08	92.83	78.36	35.91	89.98	92.51	89.85	83.69	86.05
	Zero-Mean	86.57	86.49	69.27	82.43	86.99	79.46	0.00	61.33	6.42	77.91	62.11	68.90
	Additive	**99.88**	96.76	**81.75**	98.60	95.61	70.97	32.86	85.74	86.38	90.38	83.17	86.13
	Poisson	**87.50**	**100.00**	16.36	**94.35**	**100.00**	68.93	11.12	40.24	78.81	81.52	66.37	75.54
	Salt&Pepper	92.61	93.12	81.11	91.53	**93.26**	53.43	5.06	85.98	**91.68**	84.27	76.42	80.15
	Stripes	**100.00**	**100.00**	90.44	**100.00**	**100.00**	**100.00**	41.06	70.22	**100.00**	95.11	89.08	93.63
	Deadline	84.09	76.23	64.95	91.48	90.70	90.70	1.76	48.55	91.52	75.77	71.11	66.36
	Blur	**100.00**	80.79	**96.75**	90.18	**100.00**	**100.00**	40.76	46.41	19.94	82.51	74.98	75.22
	Fog	**100.00**	**100.00**	37.90	50.94	**100.00**	52.93	7.08	15.61	8.77	75.72	52.58	65.71
SSFTT	Jpeg	70.10	91.85	61.90	63.64	60.53	67.20	60.50	64.76	59.81	77.07	66.70	69.50
	Zero-Mean	44.76	14.34	56.23	53.78	58.14	48.81	58.17	52.22	59.14	34.97	49.51	27.88
	Additive	46.36	29.56	47.41	47.19	51.59	46.73	77.59	47.04	87.68	41.29	53.46	31.98
	Poisson	17.01	15.36	17.64	17.50	17.74	17.23	17.74	17.42	17.79	16.48	17.27	10.73
	Salt&Pepper	44.42	23.11	52.46	50.74	53.79	47.26	53.82	49.65	54.50	37.56	47.75	30.01
	Stripes	41.01	28.13	45.86	44.83	46.67	42.72	46.69	44.17	47.10	36.86	43.02	26.35
	Deadline	35.95	42.23	33.58	34.09	33.19	35.11	33.18	34.41	32.98	37.96	34.97	24.86
	Blur	33.86	59.74	24.11	26.18	22.48	30.41	22.45	27.51	21.63	42.17	29.82	19.42
	Fog	81.03	96.21	75.31	76.53	74.36	79.01	74.34	77.31	73.85	85.89	78.66	81.51
Ours	Jpeg	**99.74**	**99.46**	**97.38**	**99.95**	**100.00**	**98.97**	**99.54**	**100.00**	**97.05**	**99.37**	**99.12**	**99.57**
	Zero-Mean	**95.82**	**97.70**	**75.95**	**97.91**	**100.00**	**95.07**	**90.73**	**96.14**	**34.32**	**94.35**	**87.07**	**92.39**
	Additive	**99.46**	**99.11**	78.61	**99.35**	**100.00**	**86.04**	**79.85**	**98.34**	**92.08**	**95.83**	**92.54**	**94.35**
	Poisson	85.18	94.89	**90.10**	79.88	87.36	**91.39**	**89.13**	**97.31**	**83.51**	**91.18**	**88.75**	**89.65**
	Salt&Pepper	**96.31**	**95.42**	**84.75**	**98.34**	81.79	**76.18**	**80.53**	**97.99**	80.68	**91.96**	**87.99**	**90.51**
	Stripes	**100.00**	98.52	**95.09**	99.51	**100.00**	98.45	**95.28**	**96.43**	94.63	**98.30**	**97.55**	**98.28**
	Deadline	**99.62**	**95.31**	**95.78**	**97.75**	**98.56**	**92.74**	**95.33**	**97.31**	**98.20**	**96.19**	**96.73**	**96.71**
	Blur	99.88	**99.95**	94.91	**99.96**	99.88	93.33	**92.15**	**99.71**	**73.80**	**94.65**	**90.30**	**93.17**
	Fog	86.42	99.53	**86.40**	**90.12**	**100.00**	**89.18**	**85.68**	**92.53**	**85.27**	**93.61**	**90.57**	**92.88**

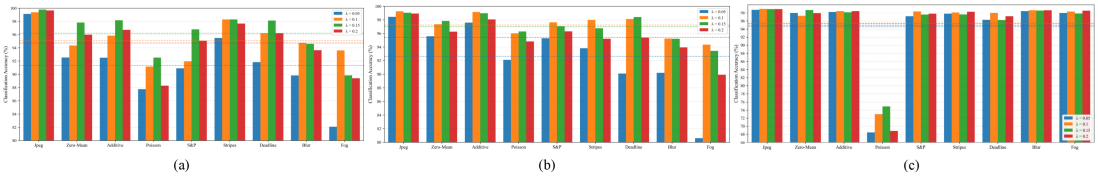


Fig. 13. Sensitivity analysis of the SupCon loss weight λ_{sup} under different degradation types. (a) PU, (b) WHLK, and (c) IP.

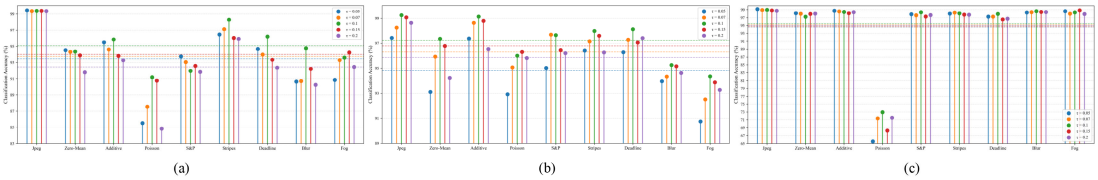


Fig. 14. Sensitivity analysis of the temperature parameter τ under different degradation types. (a) PU, (b) WHLK, and (c) IP.

to τ. Smaller temperature values lead to sharper similarity distributions but may reduce generalization, whereas larger values weaken the discriminative capability of contrastive learning.

Based on these observations, the final experiments in this article adopt $\lambda_{\text{sup}} = 0.1$ and $\tau = 0.1$ as the default parameter settings.

To investigate whether the input patch size influences the degradation interpretation and classification performance, we conduct additional experiments with different patch sizes on the PU, WHLK, and IP datasets, as shown in Fig. 15(a)–(c). The results indicate that the classification performance remains relatively stable across different degradation types when moderate patch sizes are used. Specifically, patch = 13 achieves the most balanced performance on both the PU and IP datasets, while patch = 33 provides the most stable results on the WHLK dataset, which has a higher spatial resolution and therefore benefits from larger spatial contexts. In contrast, excessively small patches may lack sufficient spatial information, whereas overly large patches may introduce redundant spatial features and increase noise interference. Based on these observations, the final experiments adopt patch = 13 for PU and IP, and patch = 33 for WHLK, which achieve a good balance between spatial context modeling and robustness to degradation.

E. Efficiency Analysis

The computational efficiency and practical feasibility of the proposed method are evaluated through comprehensive analyses of model complexity and runtime performance, including

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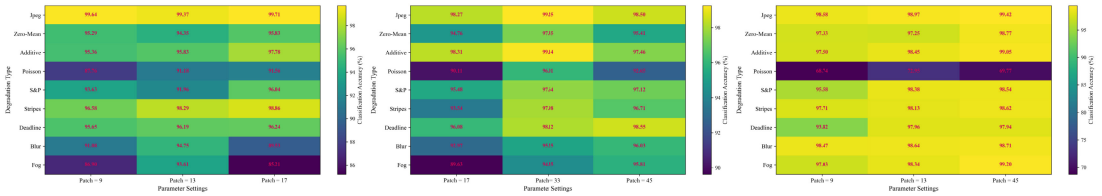


Fig. 15. Effect of input patch size on classification accuracy under different degradation types on three datasets. (a) PU, (b) WHLK, and (c) IP.

TABLE XIV
MODEL COMPLEXITY EVALUATION ON PU, WHLK, AND IP DATASETS

Dataset	Metric	HyperTTA	SSEFN	SSFTT	SCLGC	RPCL-FSL	SS-MTr	DSFormer	Ours
PU	Params(M)	0.761	12.841	0.153	**0.048**	0.072	0.725	0.677	0.435
	FLOPs(G)	0.207	6.736	**0.023**	0.224	0.174	0.033	0.060	0.101
	Time(ms)	1.204	6.854	0.809	**0.187**	0.640	1.291	6.827	1.954
	GPU Mem(GB)	0.066	0.071	0.016	0.024	**0.011**	0.012	0.014	0.014
WHLK	Params(M)	1.217	18.826	0.153	**0.031**	0.089	0.768	0.677	0.628
	FLOPs(G)	0.525	9.888	**0.023**	0.567	0.180	0.040	0.060	0.167
	Time(ms)	1.657	6.406	0.796	**0.303**	0.646	1.311	6.769	1.990
	GPU Mem(GB)	0.054	0.298	0.016	0.063	**0.011**	0.012	0.014	0.015
IP	Params(M)	0.163	16.354	0.153	**0.061**	0.082	0.677	0.751	0.548
	FLOPs(G)	0.088	8.567	**0.023**	0.104	0.178	0.060	0.037	0.139
	Time(ms)	0.347	6.749	0.870	**0.149**	0.645	6.956	1.391	1.945
	GPU Mem(GB)	**0.010**	0.159	0.016	0.013	0.011	0.014	0.012	0.014

the number of parameters, FLOPs, inference time, and GPU memory consumption. The comparisons are conducted on three benchmark datasets, namely PU, WHLK, and IP, and the detailed results are summarized in Table XIV.

As summarized in Table XIV, DIR-HSI maintains a compact model size across all datasets, with parameter counts of 0.435 M, 0.628 M, and 0.548 M on PU, WHLK, and IP, respectively. Compared with heavy transformer-based models such as SSEFN, which contains more than 12 M parameters and incurs over 6.7 G FLOPs, DIR-HSI achieves a substantial reduction in both parameter scale and computational cost. Specifically, the FLOPs of DIR-HSI are limited to 0.101 G on PU, 0.167 G on WHLK, and 0.139 G on IP, which are one to two orders of magnitude lower than those of computationally intensive baselines.

In terms of runtime efficiency, DIR-HSI demonstrates stable and competitive inference speed across different datasets, with inference times of approximately 1.95 ms per sample. Although some lightweight methods such as SSFTT and SLCGC exhibit faster inference under certain settings, they generally suffer from significant performance degradation under complex corruption scenarios, as shown in the quantitative results. In contrast, DIR-HSI achieves a favorable balance between robustness and efficiency, delivering strong degradation-invariant performance without incurring excessive computational overhead.

Moreover, DIR-HSI exhibits low GPU memory consumption, requiring only about 0.014–0.015 GB across all datasets, which is comparable to or lower than most baseline methods. This lightweight memory footprint further enhances its suitability for real-world deployment, especially in resource-constrained environments.

Collectively, the above results demonstrate that DIR-HSI not only achieves superior robustness and classification accuracy under diverse degradation conditions but also maintains moderate computational complexity and efficient inference speed. These characteristics highlight its strong potential for practical HSI analysis and engineering applications where both performance reliability and computational efficiency are critical.

IV. CONCLUSION

In this study, we addressed one of the fundamental challenges in HSI classification—achieving robust and discriminative representation learning under complex and multisource degradations. Conventional spatial–spectral models primarily focus on spatial or spectral noise suppression but neglect the intrinsic statistical and structural characteristics of degradation in the frequency domain, leading to unstable feature representations and domain shifts across diverse degradation types. To overcome these limitations, we proposed DIR-HSI, a unified degradation-invariant classification framework that systematically integrates frequency-domain modeling and discriminative contrastive optimization. Specifically, the DIR-FFE module performs Fourier-domain decomposition and adaptive spectral suppression to disentangle degradation-sensitive and stable components, while the DIR-CE module introduces cross-degradation SupCon learning to align same-class embeddings and enlarge interclass separability. These two modules collaboratively enable the model to achieve both degradation invariance and feature discriminability, significantly enhancing robustness and generalization in complex degradation scenarios. Extensive experiments on the PU, WHLK, and IP datasets demonstrated that DIR-HSI consistently outperforms state-of-the-art methods in accuracy, stability, and cross-degradation generalization, validating the effectiveness of the proposed degradation-invariant representation mechanism.

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Despite these promising results, several limitations of the proposed framework remain. First, although DIR-HSI incorporates dynamic frequency-aware mechanisms through learnable fusion and adaptive frequency selection, the frequency modeling is still constrained to a limited set of predefined filtering operators, which may restrict its expressiveness when encountering highly irregular or previously unseen degradation patterns. Second, the current study focuses on robustness against single degradation types, whereas real-world hyperspectral remote sensing data often suffer from compound or mixed degradations arising from multiple sources simultaneously. Effectively modeling and disentangling such complex degradation interactions remains a challenging problem and deserves further investigation. In addition, the proposed framework is evaluated under patch-based classification settings, and its scalability to extremely large-scale scenes or real-time onboard processing scenarios warrants further exploration.

Looking forward, future research can extend DIR-HSI along several meaningful directions. One potential avenue is to investigate degradation-aware feature disentanglement under compound corruption scenarios, enabling more explicit separation of overlapping degradation effects in both spatial and frequency domains. Another promising direction is to explore the integration of DIR-HSI with large-scale pretraining or foundation-model paradigms to enhance representation transferability across sensors, scenes, and acquisition conditions. Furthermore, extending the framework toward efficient full-scene inference and deployment-oriented optimization may facilitate its practical adoption in real-world hyperspectral analysis systems.

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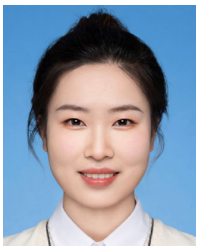
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